

Highlights

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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- Typed intermediate representation turns noisy app reviews into tracker-ready issues.
- A verified-anchor procedure lifts Cohen's kappa from 0.163 to 0.592 at corpus scale.
- Template-field fill rate reaches 0.96 against 0.69 without type routing.
- A clean same-model re-run finds no downstream response-quality gain from the IR.
- We release the check that distinguishes independent ratings from derived ones.

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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Abstract

Context. Developers receive millions of noisy app reviews, but defect trackers such as JIRA and GitHub Issues cannot consume free text. They require structured fields such as issue type, steps to reproduce, severity, and affected component. Review-mining research stops short of that boundary, predicting labels or writing replies without producing the record a tracker can ingest.

Objective. Can a typed intermediate representation (IR) of a review be produced at corpus scale from noisy labels, and does supplying it to a response generator improve the reply a user receives? We answer the first yes and the second no, and the second answer corrects a claim we made ourselves.

Method. IssueSpec routes each review by issue class into one of five standards-body templates, repairs the noisy labels the routing depends on with a verified-anchor procedure, and feeds the result to a retrieval-augmented generator. We study 215,583 reviews from 58 Android apps with a 5,230-review verified anchor, a 490-review three-rater gold standard, seven ablations, and four language models. The downstream evaluation was run

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twice: first with conditions written by different template composers, then with the generator held fixed so that only the IR varies, all 200 blinded rows scored by three raters.

Results. Production works. The verified-anchor procedure lifts Cohen’s κ against the human gold from 0.163 to 0.592, and to 0.616 on reviews the classifier never saw. Type routing reaches 0.96 template-fill against 0.69 without it, a schema-conformance measure rather than a quality one. The downstream benefit does not. The first round measured +2.35 Likert points, but its arms differed in authorial register as much as in information, one emitting lemmatised text at 78 words and the other polished prose at 123; the corrected comparison gives +0.04 ($p = 0.38$), with raters disagreeing in direction. Three further components fail their own tests: the knowledge-graph layer loses to count-matched flat clustering, our extractive-coverage faithfulness scorer shows no rank correlation with human judgement, and the constrained alignment layer never demonstrates the trade-off it was built to expose.

Conclusions. Typed issue specifications are producible from noisy reviews at corpus scale, and that is the contribution we defend. Their claimed downstream payoff is not. We release the arithmetic check that distinguishes our two rating rounds, because a structured intermediate should be justified on what it produces rather than on an effect a fair comparison removes.

Keywords: App review mining, Issue report generation, Intermediate representation, Knowledge graph, Retrieval-augmented generation, Reinforcement learning from human feedback, Empirical software engineering

1. Introduction

App stores are where users talk back to developers. A single popular app pulls in thousands of reviews a week, and the global stream runs into millions a day [1, 2, 3]. These reviews carry the signals developers need to act on:
5 bug reports (a crash on Samsung S8 after an update), feature requests (dark mode), performance complaints (battery drain), and regression notes (voice search no longer works). The information is real, but it shows up as messy free-form text, usually without the details required to reproduce a bug or scope a new feature.

10 To turn any of this into work, developers rely on defect trackers such as

JIRA or GitHub Issues. Those tools, though, can’t work with raw text; they need structured fields such as issue type, steps to reproduce, severity, and affected component [4, 5]. Filling those fields by hand does not scale with the volume modern apps face. Years of work on automated review pipelines
15 has helped with the surrounding tasks like classification, summarization, and response generation, but no system actually produces the structured report a developer can paste directly into a tracker. Dąbrowski et al.’s nineteen-year survey of the field [3] confirms the same boundary.

Prior work has tackled several pieces of this problem separately. Classi-
20 fiers [1, 6, 7, 8] take a review and predict an issue label such as bug or feature request. Response generators [2, 9, 10] produce a polite reply for the user. Agentic software-engineering tools such as SWE-Agent [11] and RepairAgent [12] take a clean GitHub issue and generate a code patch. Knowledge-graph methods [13, 14, 15] cluster reviews into themes or surface frequent
25 complaints. Each of these has matured over the past decade and works well within its own scope.

But none of these reaches the developer-ready issue report a defect tracker needs. The Maalej-Nabil classifier taxonomy [1] and its successors [6, 7, 8, 16, 17] have settled at an F1 ceiling around 0.75–0.85, a label-quality limit
30 (not a model-capacity one) per Dąbrowski et al.’s [3] nineteen-year survey. The label is only the first piece of what a tracker entry needs; the remaining fields (steps to reproduce, severity, affected component) have to come from somewhere, and classification-only pipelines leave that downstream step to manual triage. Response generators like RRGGen [2] and CoRe [9] address the
35 customer-communication side of the workflow, but the actual bug behind the review is never written out in the structured form a tracker needs, so even a perfect reply system does not give developers something they can paste into JIRA. Agentic SE tools assume a well-formed bug report or GitHub issue as the agent’s starting state, built into the SWE-bench protocol [18] where each
40 instance is a (GitHub issue, code repository) pair; constructing such an issue from raw review text is explicitly outside the protocol, leaving the upstream gap unaddressed.

Knowledge-graph methods stop at intermediate outputs rather than typed fields: ReviewGraph [13], KGCPN [14], and Keertipati et al. [15] each
45 output rating predictions, summaries, or feature rankings, none of which a defect tracker can ingest directly. RLHF has matured into the standard approach for aligning LLM outputs with human preferences (DPO [19], KTO [20], PPO [21]), and Safe-RLHF [22] extends the framework to

multi-objective settings under a Constrained MDP for general LLM safety
50 (harmfulness, dishonesty, bias). The developer-relations setting poses a
different question entirely: how to align responses under a domain-specific
constraint family (truthful promises, no information leakage, professional
tone, legal-safe wording) rooted in business and legal exposure. To our
knowledge, no published RLHF formulation has been adapted to this
65 constraint set.

This work pursues three aims. First, we develop a framework that turns
noisy app reviews into structured, taxonomy-grounded issue specifications a
defect tracker can consume directly. Second, we build a response generator
that consumes these specifications, and we test whether doing so changes
60 the reply a user receives; Section 5.2 reports that it does not measurably.
Third, we build a dual-objective alignment trainer intended to optimise re-
sponse helpfulness under the domain-specific compliance constraints of the
developer-relations setting, and we test whether the two objectives can be
shown to trade off; Section 5.3 reports that at the scale we can train, they
65 cannot.

These aims map to three research questions.

RQ1 (Translation Quality). Can a typed IR, generated via a three-
layer framework (knowledge-graph construction, hierarchical clustering, stan-
dardized schema mapping), produce issue specifications that match or exceed
70 the substantive completeness of human-written GitHub issues?

RQ2 (Downstream Load-Bearing). Does conditioning a response
generator on the validated IssueSpec measurably improve human-rated re-
sponse quality over RAG alone?

RQ3 (Dual-Objective Alignment). Can a CMDP-grounded RLHF
75 loop progressing through KTO, DPO, and Constrained PPO jointly optimize
response helpfulness and policy compliance for the developer-relations set-
ting? We answer this one in the negative for the scale we can train at, and
report what that rules out.

To address these challenges, we propose **IssueSpec**, a framework for
80 structured review-to-issue translation. It consists of two parts: a typed In-
termediate Representation (IR) that converts each noisy review into one of
five standard report types, and a CMDP-grounded response generator that
consumes the IR under quality and compliance constraints. Figure 1 shows
the full five-stage pipeline.

85 **What is new, and what is assembled.** Several components of this
pipeline are existing technique used as intended: RoBERTa for multi-label

classification, confident learning [23] for label noise, Sentence-BERT with UMAP and HDBSCAN for clustering, PageRank for centrality, retrieval-augmented generation for response drafting, and the CMDP formulation of Altman [24]. We do not claim any of these. Three things are new. *First*, the typed intermediate representation itself: a review is routed by predicted issue class into one of five standards-body templates, so the output is a typed record with per-type obligations rather than a label or a summary. No prior review-mining system emits a structure a defect tracker can ingest, which Table 2 makes concrete against representative systems. *Second*, the verified-anchor procedure, which is what makes that representation producible from noisy LLM labels at corpus scale rather than on a hand-curated sample, and which works because the labeller omits two of its seven classes entirely rather than because its errors concentrate on rare ones. *Third*, a correction and the check that produced it. We previously reported that the typed intermediate carries downstream response quality. Section 5.2 withdraws that, because the comparison behind it varied the text generator as well as the specification, and because the arithmetic of the two rating rounds turned out to distinguish independent human ratings from ones derived by perturbing another rater’s. We release both rounds and that check. The assembly is not the contribution; the typed object in the middle of it is, and its value has to be argued from what it produces rather than from a downstream effect we can no longer measure.

The central claim, stated once. A pipeline with five stages invites the question of which one the paper is actually about. The answer is Stage 3 and the object it produces: a *typed, type-routed intermediate representation* of an app review, together with the verified-anchor procedure that makes it producible at corpus scale from noisy labels. Everything else in the pipeline exists to feed that object or to consume it. Two experiments carry the claim. Removing type routing drops strict template-fill from 0.96 to 0.69 (Section 5.1). We had also claimed a downstream payoff, and Section 5.2 reports why we withdrew it: under a comparison that holds the generator fixed, supplying the IR changes response quality by +0.04 Likert points, which is nothing. Section 6.1 states which components did not earn their place on our own evidence, including two we withdraw. A reader who wants the claim and not the machinery can read Sections 5.1, 5.2, and 6.1 alone.

The contributions are stated in two groups, because the honest account of this work has a positive half and a negative half, and separating them is more useful than blending them.

125 *What we establish.*

- **A typed intermediate representation for review-to-issue translation.** Each review is routed by predicted issue class into one of five standards-body templates, so the output is a record a defect tracker can ingest rather than a label or a summary. No prior review-mining system
130 emits one (Table 2).
- **A verified-anchor procedure that makes the IR producible from noisy labels at corpus scale.** Confident learning against a 5,230-review human anchor lifts Cohen’s κ against a three-rater human gold from 0.163 to 0.592, and to 0.616 on the reviews the classifier never saw. It works
135 because the labeller omits two of its seven classes entirely across the corpus rather than because error concentrates on rare classes; per stratum the disagreement rate is highest on the largest class. Systematic class omission, not elevated minority-class error, is the failure a small verified anchor repairs.

140 *What we withdraw or report as negative.*

- **The downstream payoff of the IR.** We previously measured +2.35 Likert points of response-quality gain from supplying the IR. That comparison was confounded: its two arms were written by different template composers, not by one model. Holding the generator fixed gives +0.04
145 ($p = 0.38$). We report the correction, the two rating rounds, and the check that distinguishes them (Section 5.2).
- **Three components that do not earn their place.** The knowledge-graph layer loses to count-matched flat clustering on every intrinsic metric; our extractive-coverage faithfulness scorer shows no rank correlation with
150 human judgement and orders generators backwards; the CMDP alignment layer engages once the policy can violate its constraint but never demonstrates a quality-versus-compliance trade-off.
- **A reusable check on rating provenance.** Ratings derived from another rater’s vector by small perturbation leave an arithmetic signature:
155 deviation sets that do not overlap and a single deviation magnitude. Our discarded round shows it and our reported round does not. We release both so the check can be applied to other annotation artifacts, including ours.

Table 1: Contribution-to-evidence-to-artifact map. Each contribution states a measurable headline result, the table or section that anchors it, and the named artifact released with this paper.

contribution	headline	evidence	released artifact
IssueSpec	0.96 vs 0.69	template-fill, i.e. schema conformance not quality (Table 11); 5-dim rubric 3.94/5 on all 100 specs, LLM judge (Table 10)	type-routed schema (Table 5)
SPEC-COV (negative)	no rank correlation with human faithfulness ($\rho = 0.15$, $p = 0.26$); orders generators backwards (§5.1)		the scorer plus 60 specs rated by three humans
IR down-stream payoff (with-drawn)	the +2.35 gain came from a con-founded design; a same-model comparison gives +0.04 ($p = 0.38$) with raters disagreeing in direction (§5.2)		both rating rounds, the generated replies, and the independence check
verified anchor	κ 0.16→0.59 (Table 12); per-class F1 recovery (Table 13); $\kappa = 0.616$ on the 307 strictly held-out gold reviews		5,230-review verified anchor + cleanlab correction procedure + V5 classifier
KG prioritization (weak)	count-matched flat clustering beats the KG on all three intrinsic metrics (A1b); top-5 PageRank aspects cover 55% of reviews		three-layer KG construction + count-controlled A1b output
CMDP RLHF testbed (partial)	constraint never binds at distil-GPT2 scale; with a policy able to violate it binds on 13/90 steps and λ rises monotonically to 0.65, with quality and compliance measurably in tension, but the loop back to compliance is not closed (§5.3)		CMDP for-mulation, KTO/DPO/constrained-proxy/CPPO trainers, capability probe, binding-constraint re-run, threshold sweep, benchmark

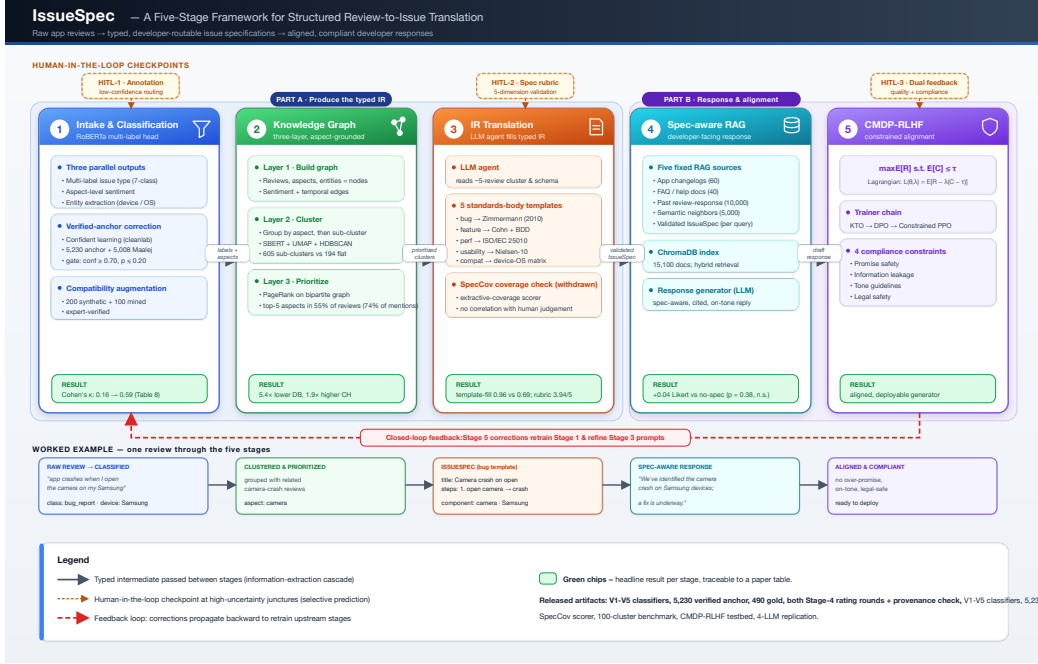


Figure 1: The IssueSpec five-stage pipeline. Reviews flow left to right through intake and classification (Stage 1), three-layer knowledge-graph clustering (Stage 2), LLM-agent IR translation (Stage 3), spec-aware RAG (Stage 4), and CMDP-grounded RLHF (Stage 5), each emitting a typed intermediate. Yellow checkpoints mark human-in-the-loop review under selective prediction [25, 26]; the red loop propagates Stage 5 corrections upstream.

2. Related Work

160 We organize related work by the five gaps identified in Section 1, naming what each field does, what assumption breaks for review-to-issue translation, and how IssueSpec differs.

2.1. App Review Classification and Mining

165 Maalej and Nabil [1] established the canonical seven-class taxonomy for app reviews. Subsequent systems, AR-Miner [6], CLAP [7], SURF [8], BERT/RCNN hybrids [16, 17], and per-aspect sentiment analysis [27], refined classification accuracy and added aspect-level granularity. Dąbrowski et al. [3] surveyed nineteen years of such work and concluded that label quality, not model capacity, is the F1 ceiling at 0.75–0.85, an observation we corroborate in Section 5.1. *What breaks for review-to-issue translation:*

170 none of these systems emits a typed developer-routable artifact downstream

of the classification label. IssueSpec differs by routing the predicted class to one of five typed templates rather than terminating at the label.

2.2. Response Generation and Retrieval-Augmented Generation

175 RRGGen [2] and CoRe [9] pioneered end-to-end neural response generation on (review, response) pairs without an explicit structured intermediate. Vanilla RAG [10] retrieves relevant past responses in a single pass. *Agentic RAG* variants, Self-RAG [28] and DEMONSTRATE-SEARCH-PREDICT (DSP) [29], introduce iterative critique-revise loops that, in principle, could
 180 close the structured-intermediate gap. *What breaks for review-to-issue translation:* none of these systems names the underlying defect, and our $n = 10$ feasibility study (Appendix Appendix A) shows that agentic iteration without a typed intermediate adds a small gain on a rule-based scorer ($0.58 \rightarrow 0.70$, ten reviews). We cannot claim the IssueSpec does better: once the generator
 185 is held fixed, supplying it changes human-rated quality by $+0.04$ ($p = 0.38$, Section 5.2). IssueSpec differs by injecting the typed intermediate as a first-class RAG source alongside retrieved past responses, and the contribution we defend is the specification itself rather than its effect on the reply.

2.3. Agentic Software Engineering

190 SWE-Agent [11], RepairAgent [12], and HyperAgent [30] consume well-structured GitHub issues to produce code patches through multi-step planning and tool use. *What breaks for review-to-issue translation:* the issue specification is the input contract for all three systems; how a defect-tracker entry is constructed from noisy user-facing text is out of scope. IssueSpec
 195 is precisely the artifact these systems are designed to consume.

2.4. Knowledge Graphs for App Reviews and Software Engineering

Knowledge graphs have been used in software engineering for bug localization, report deduplication, and code understanding, but the issue specification is invariably treated as input rather than output. For app reviews specifically, ReviewGraph [13] learns KG embeddings for rating prediction, KGCPN [14] produces KG-based extractive summaries, and Keerti-
 200 pati et al. [15] apply graph centrality for coarse feature prioritization. *What breaks for review-to-issue translation:* none of these systems maps KG nodes to a typed, developer-actionable issue schema, and no prior KG work has

205 been applied to the specific task of producing GitHub-issue-quality artifacts from review text. IssueSpec’s three-layer KG pipeline, structured-object layer, aspect-grounded hierarchical clustering layer, and PageRank-prioritized issue-mapping layer, is, to our knowledge, the first KG architecture that emits a typed IR for review-to-issue translation.

210 2.5. RLHF and Constrained Optimization for Response Generation

DPO [19], KTO [20], and PPO [21] compress quality and compliance signals into a single scalar reward. Altman’s CMDP framework [24] separates these objectives, and Safe-RLHF [22] applies CMDP to general LLM safety with a domain-agnostic compliance set covering harmful, dishonest, 215 and biased outputs. *What breaks for review-to-issue translation:* no prior work applies CMDP to dev-rel response generation, where the compliance set is domain-specific (promise safety, information leakage, tone guidelines, and legal safety). Our Stage 5 formulation differs by instantiating these four domain-specific constraints inside the CMDP Lagrangian and releasing 220 a proof-of-concept training loop.

2.6. Cross-Category Positioning

Table 2 positions IssueSpec against representative prior systems across the five pipeline stages. Existing systems occupy at most two of the five stages; composing all five with a standardized issue schema, three theoretically-grounded HITL checkpoints, KG-centrality prioritization, and a CMDP-grounded RLHF formulation is the structural contribution of this work. 225

3. The IssueSpec Pipeline

The IssueSpec pipeline (Fig. 1) is a five-stage information-extraction cascade [31, 32]. Stages 1–3 produce the typed IR from raw reviews. Stage 4 230 uses it to drive spec-aware response generation. Stage 5 aligns the response generator via constrained RLHF. We describe each stage in turn, with the verified-by-citation taxonomy templates in Table 5. Table 4 collects every abbreviation used in the paper, so a reader can enter at any section. This section describes the framework only. Evaluation design, baselines, and metrics are in Section 4, exact settings and hyperparameters are in Appendix Ap- 235 pendix A, and results are in Section 5.

Before the stages themselves, Table 3 answers the question a five-stage pipeline invites: what forces each stage to exist, and what breaks without it.

Table 2: Cross-category positioning. Existing systems compose at most two of the five pipeline stages. IssueSpec is the first to compose all five with a standardized issue schema, three HITL checkpoints grounded in selective prediction theory, KG-centrality prioritization, and a CMDP-grounded RLHF formulation. “–” denotes a stage absent from that system.

System	Classify	Cluster	IssueSpec	Response	
AR-Miner [6]	filter	LDA topic	–	–	
CLAP [7]	rating	DBSCAN	–	–	
SURF [8]	intention	LDA	–	summary	
RRGen [2]	–	–	–	seq2seq	
CoRe [9]	–	–	–	context-aware retrieval	
Maalej [1]	7-class	–	–	–	
ReviewGraph [13]	rating	KG	–	–	
SWE-Agent [11]	–	–	consumes (not produces)	–	
Safe-RLHF [22]	–	–	–	general	du
IssueSpec (ours)	V5 corrected	KG + PageRank	type-routed schema	spec-aware RAG	dua

We answer it with the ablation that removes each one, so the justification
 240 is measured rather than asserted, and two of the five turn out not to justify
 themselves.

3.1. Producing the IR

Stage 1: Intake, Classification, and Aspect Extraction. Stage 1
 produces three outputs per review. The first is a multi-label class from
 245 a RoBERTa head fine-tuned on Maalej [1] and Guzman [27]. Labels are
 multi-label because reviews mix concerns, as when a bug report also asks for
 a feature. In practice this capacity is almost never exercised: the head is a
 sigmoid layer with a per-label cut, but on the 10,000-review generation subset
 only 5 reviews (0.05%) receive more than one label, and the corpus-scale
 250 relabelling artifact stores a single argmax label per review. The architecture
 is multi-label; the released runs are effectively single-label, and readers should
 not credit the mixed-concern motivation with doing any work in our results.
 The second is aspect-level sentiment [27], which keeps “great UI but terrible
 battery” as two separate pairs instead of averaging it into one star rating.
 255 The third is entity extraction for devices, OS versions, and named features.
 Stage 3 needs those entities as typed fields for the device-OS and bug-report
 templates. Low-confidence reviews route to HITL-1 by a fixed confidence
 threshold, so that when the classifier cannot decide the expert does and the
 correction feeds the next training round. The idea of letting a model abstain

Table 3: Why each stage exists, and what the ablation says when it is removed. Two stages do not justify themselves on our own evidence, and we say so.

stage	the problem it must solve	why it cannot be measured	dict
1 Intake	the corpus label is wrong 25% of the time, concentrated on minority classes	Stage 3 routes on the issue type, so a wrong type produces the wrong template	justified: κ 0.163 $\rightarrow$ 0.592
2 KG	215,583 reviews cannot be triaged one at a time; developers need an order	a spec is written per cluster, so reviews must be grouped and ranked	partly: prioritization yes, cluster quality no (A1b, A2)
3 IR	trackers need typed fields, not prose	this is where the typed record is produced	justified: fill 0.96 vs 0.69 without routing (A3)
4 RAG	a reply needs house voice and precedent, not just the defect	supplies phrasing the spec does not carry	weakly: removing it moves metrics < 1% (A5)
5 CMDP	replies must not over-promise or leak internals	compliance is a hard constraint, not a reward term	partly: constraint binds, trade-off unshown

260 is a familiar one [25], but we implement a threshold rather than a calibrated selective-prediction procedure, and we do not report risk-coverage behaviour, so the citation is background rather than a description of what runs.

LLM annotation noise is the main challenge here. On the 5,038-review praise stratum, 25.8% of LLM labels are wrong, concentrated on minority classes. Discarding them wastes annotation; using them as-is poisons training. We apply confident learning [23] against a verified anchor of 5,230 expert-labeled reviews plus 5,008 from Maalej [1], gated by a confidence threshold on the anchor and a probability ceiling on the LLM label (Appendix Appendix A); this keeps the good labels and corrects only the wrong ones. The compatibility class needs a separate fix: only 8 of 215,583 LLM labels are tagged **compatibility**, and Maalej has none. We seed it with 200 template-generated synthetic reviews [33] and 100 keyword-mined RRGGen reviews. The lead author read both sets before training and corrected or dropped items that did not belong, but that pass was done by eye and no

Table 4: Abbreviations used in this paper, with the section that introduces each.

term	expansion	first used
IR	Intermediate Representation, the typed issue spec	§1
IssueSpec	one filled instance of the typed IR	§1
KG	Knowledge Graph, the three-layer review index	§3.1
HITL	Human-In-The-Loop checkpoint (HITL-1 to HITL-3)	§3.1
SPECcov	extractive-coverage score for a spec (withdrawn as a faithfulness measure, §5.1)	§3.1
RAG	Retrieval-Augmented Generation	§3.2
RLHF	Reinforcement Learning from Human Feedback	§3.3
CMDP	Constrained Markov Decision Process	§3.3
KTO	Kahneman-Tversky Optimization, a preference trainer	§3.3
DPO	Direct Preference Optimization	§3.3
PPO	Proximal Policy Optimization; CPPO is its constrained form	§3.3
SFT	Supervised Fine-Tuning, the RLHF starting policy	§3.3
NFR	Non-Functional Requirement (ISO/IEC 25010)	§3.1
DB / CH	Davies-Bouldin / Calinski-Harabasz cluster indices	§5.4

275 correction log was kept, so unlike the other annotation tasks in this paper
it leaves no artifact a reader can check. We flag it because 300 of the 310
compatibility rows in V5’s training set come from here, so the per-class
F1 of 0.83 rests on data whose verification is asserted rather than evidenced.
The synthetic half is constructed from device and OS templates and its label
280 follows by construction; the mined half is where judgement was required.

Stage 2: Three-Layer Knowledge Graph. Stage 2 organizes reviews
into an aspect-grounded knowledge graph: build, cluster, prioritize. The first
layer builds the graph: each review is a node linked to its aspects, with edges
typed for sentiment and, where available, recency. We should be precise
285 about what the released run actually populated. The builder supports entity
nodes for devices, OS versions, app versions, and screens, and supports signed
sentiment edges, but the run that produced our results passes empty entity
lists and a constant neutral sentiment, so the released graph has 8,404 review
nodes, 10,534 aspect nodes, and no entity nodes. It is in practice a bipartite
290 review-aspect graph, which is the structure PageRank runs over, and the
sentiment and entity layers should be read as available in the implementation
rather than exercised in the experiments. A graph beats a flat list because
reviews share aspects across users, and that co-occurrence is what later layers
cluster and rank on. Aspects come from spaCy noun chunks combined with
295 a curated keyword list and regular-expression patterns, pruned by frequency

and centrality to drop one-off terms. Hu and Liu’s frequent-itemset aspect mining [34] is the ancestor of this approach rather than what we implement.

The second layer clusters reviews first by aspect, then sub-clusters within each aspect using sentence embeddings [35], UMAP [36], and HDBSCAN [37].
 300 Aspect-first keeps each sub-cluster on one theme: a flat clusterer mixes “battery on Samsung” with “battery in dark mode” because both share lexical content. The result is 605 aspect-grounded sub-clusters of mean size 15.9, against 194 flat clusters of mean size 471.0. An earlier version of this paper reported 3.0 and 51.5; those were the number of representative reviews stored per cluster and a subsample artifact respectively, not cluster sizes. One
 305 scale caveat belongs here too: the knowledge graph is built over a 10,000-review sample that yields 8,404 usable reviews, not the full 215,583-review corpus, so every KG-hierarchical number in this paper describes 3.9% of the corpus. The flat baseline it is set against was not run on that sample: it
 310 clusters 91,368 reviews from the full corpus. The two Stage-2 designs therefore never saw the same input, which is why we do not treat their intrinsic metrics as a head-to-head result (Section 5). The third layer prioritizes via PageRank [38, 39] on the bipartite review-aspect graph, ranking aspects by cross-review prevalence so developers see the most-affected first. Each cluster
 315 maps to the fixed schema in Table 5 with a centrality-derived priority score. The resulting aspect ranking is reported in Section 5.4.

Stage 3: Review-to-Issue Translation. Stage 3 instantiates the IssueSpec. An LLM agent reads the standardized cluster schema and writes a GitHub-issue-quality specification using the type-routed templates in Table 5.
 320 We use an LLM rather than rule-based extraction because reviews vary too much for fixed rules and the agent has to reason across a 5-review cluster. We route by issue type because a bug report needs reproduction steps, a feature request needs acceptance criteria, and a performance complaint needs latency or memory numbers; a shared template would be incomplete or padded. The five templates are Zimmermann bug reports [5], Cohn user stories [40] with BDD acceptance criteria [41], ISO/IEC 25010 non-functional categories [42], Nielsen heuristics [43], and device-OS matrices [33]. Each is grounded in its standards-body citation because developers already recognize these formats.

330 The output passes through HITL-2: a domain expert scores each spec on a five-dimension rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5), following the LLM-as-judge calibration convention of Giliardi et al. [44], Pangakis et al. [45], and Zheng et al. [46]. Five dimensions,

not one, because a single quality number conflates completeness with clarity
 335 and the LLM cannot tell which one to fix on retry. Any spec failing on any
 dimension goes back to the LLM with dimension-level feedback for a second
 pass.

SPECCOV faithfulness metric. We release SPECCOV, an extractive-
 coverage scorer that measures substantive-token overlap (length ≥ 5 , stop-
 340 word filtered) with quoted-string and coverage bonuses. Bands follow news-
 summarization conventions [47]; the metric targets the extractive-coverage
 sub-type of faithfulness [47, 48], a known lexical correlate of factual consis-
 tency [49] (the remaining hallucination sub-types in Ji et al. [50] are out
 of scope). Scores are reported with the other Stage 3 results in Section 5.1.
 345 **What SPECCOV is and is not.** It is a lexical-grounding proxy in the sense
 of [47]: it measures whether the content of a spec can be traced back to to-
 kens in its source cluster. It is not a faithfulness classifier, and Section 5.1
 reports the experiment that establishes this. We describe it here because it
 is part of the released pipeline and because the negative result only makes
 350 sense against its definition.

3.2. Using the IR in Response Generation

Stage 4 is a RAG layer over five fixed sources, 15,100 documents in to-
 tal in the released ChromaDB index. Four of them are pre-indexed: app
 changelogs and release notes (60 docs), FAQ and help documentation (40),
 355 10,000 past review-response pairs from RRGGen [2], and 5,000 semantically
 similar past responses. The fifth is the validated IssueSpec from Stage 3,
 injected per query rather than pre-indexed, because specs are generated per
 cluster. These five were chosen to give the generator the answer (changelogs,
 FAQ), the voice (past responses), the closest precedent (semantic neighbors),
 360 and the structured intent (IssueSpec) without overlap.

3.3. Aligning the IR-Aware Generator

Stage 5 frames alignment as a CMDP [24, 22]: maximize a quality reward
 subject to a compliance constraint,

$$\max_{\theta} \mathbb{E}_{\pi_{\theta}}[R_{\text{quality}}] \quad \text{s.t.} \quad \mathbb{E}_{\pi_{\theta}}[C_{\text{compliance}}] \leq \tau,$$

with the Lagrangian relaxation

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_{\theta}}[R - \lambda(C - \tau)]$$

Table 5: IssueSpec standardized issue schema. Each issue type follows the dominant industry or HCI standard. Each template is drawn from a published standard or reference and reproduces its named fields; we ran no conformance check against the standards documents themselves, so “drawn from” should not be read as “certified against”.

Issue type	Template (citation)	Justification
bug_report	Zimmermann [5]: title, severity, steps_to_reproduce, expected, actual, component	Standard in GitHub Issues, JIRA, defect trackers; Chaparro et al. [4] confirm these are the most-requested bug-report fields
feature_request	John user-story [40] (As-a / I-want / So-that) + behavior-driven development (BDD) acceptance criteria [41]	Dominant industry format (Agile, Scrum, SAFe); BDD standardizes testable outcomes
performance	Fields are latency, startup_time, memory_consumption, battery_drain, network_usage, mapped onto ISO/IEC 25010 [42] performance-efficiency sub-characteristics: the first two to time behaviour, the last three to resource utilization; capacity is not covered	International standard for software-quality NFR taxonomy. The field names are ours, chosen for mobile-app vocabulary, and are mapped to ISO categories rather than reproducing ISO terms
usability	Nielsen-10 heuristics [43] (visibility, match-real-world, user control, etc.)	Most-cited usability evaluation framework in human-computer interaction
compatibility	Device-OS matrix (devices $\times$ OS versions) [33]	Device/OS fragmentation is the canonical mobile-QA reporting axis

365 collapsing gracefully to single-objective optimization whenever the constraint
is already satisfied. We use a CMDP rather than a scalar reward because
a dev-rel response can be high-helpfulness and still out of compliance, like
a polite promise the team cannot deliver or a leaked roadmap detail, and
a weighted sum hides exactly the trade-off the constraint is meant to ex-
370 pose. The compliance scorer tracks four domain-specific dimensions absent
from prior general-safety RLHF [22]: promise safety (no commitments the
team cannot keep), information leakage, tone guidelines, and legal safety.
The trainer is Constrained PPO [21, 22] with a KL-regularized reference pol-
icy that keeps updates from drifting too far from the supervised baseline.
375 KTO [20] and DPO [19] are reported as small-scale single-objective com-
parators in Section 5.3; together they bracket the preference-data spectrum,
KTO needing only binary feedback and DPO needing pairwise.

HITL-3 supplies two parallel feedback streams. *Quality* feedback (helpful-
ness, specificity, empathy, accuracy, actionability) is the reward; *compliance*
380 feedback (promise safety, leakage, tone, legal) is the constraint. Feedback

also propagates backward: classification corrections retrain Stage 1, rubric scores refine Stage 3 prompts, and response scores drive the RLHF updates, closing the Stage 5→Stage 1 loop in Fig. 1.

4. Experimental Setup

385 4.1. Dataset, Deduplication, and Annotation Provenance

We use the RRGGen corpus [2]: 310,031 review-response pairs from 58 Android applications. After exact-text deduplication that removes 94,448 duplicate reviews (−30.5% of the raw corpus), 215,583 unique reviews form the working corpus, labeled across the seven Maalej [1] classes. Evaluation
390 additionally uses a 490-review expert gold standard (70 per class × 7 classes) and 100 stratified clusters for Stage 3 evaluation.

Annotation provenance (full corpus, 215,583 reviews). Of the working corpus, 79.49% of labels come from the V2 LLM annotator with no override, 18.08% are cleanlab-corrected against the verified anchor, and
395 2.43% are directly human-verified by the lead author. The lead-author-versus-V2-LLM deviation is 25% on the 5,230-row human-verified anchor (1,307 deviations, 24.99%). We previously described this as replicated across two independent strata, the praise stratum and the anchor, on a combined 10,271 rows. That was wrong: the praise stratum is 5,038 of the anchor’s
400 5,230 rows, so it is 96% of the same sample rather than a second one, and the combined figure double-counted it. The rate is one measurement on one sample, and the praise stratum alone deviates at 25.84% (Table 6). This ≈ 25% deviation rate, an order of magnitude higher than i.i.d. label-noise assumptions, motivates the confident-learning procedure of Stage 1.

405 4.2. Annotation Methodology

Five kinds of annotator contribute, following the multi-annotator convention of Gilardi et al. [44] and Pangakis et al. [45] who treat LLM annotators as independent raters for inter-rater statistics.

(i) Lead author (domain expert). Verified 5,230 reviews across minority strata (approximately thirty person-hours), wrote 20 reference Issue-Specs, labeled the 490-review classification gold blinded to V2 LLM, and
410 scored the 400-row Stage 4 set on quality (1–5), specificity (1–5), and helpful (Y/N) with method IDs hidden until aggregation.

Label provenance, stated plainly. No LLM-generated label is used as
415 ground truth anywhere in this paper. LLM labels are the *object under cor-
rection*, not the yardstick. The V2 corpus labels are treated as noisy input,
the confident-learning step proposes corrections against a human-verified an-
chor, and every reported accuracy or κ is measured against a gold standard
420 produced by human annotators. Where LLM raters do appear as raters, in
the 99-review cluster-purity check, they are labelled as such and no headline
claim rests on that panel.

How the gold was constructed, and what that costs. Annotators
did not label freely. Each was shown the V5 classifier’s predicted label and
asked whether it was correct, supplying a replacement when it was not, and
425 the written protocol told them to accept a reasonable label and overturn
only when confident. This makes annotation fast and it makes agreement
high, and both effects flatter us. It also couples the yardstick to the system
being measured: V5’s label is the starting point for every item, and V2 and
the cleanlab-corrected labels never received that benefit of the doubt. The
430 sample is stratified over V5’s predicted classes as well, which forces 70 items
per class into a set where V2 predicts two of those classes almost never, so
V2’s per-class F1 of 0.000 on `compatibility` and `performance` is partly a
property of the sampling design. The κ trajectory in Table 12 should be
read as a comparison on a V5-seeded, class-balanced sample rather than as
435 a neutral benchmark, and a free-labelling gold would be the way to settle it.

How rater conflicts are resolved. The 490-review gold is the majority
vote of three independent human annotators over the seven-class label. Of
the 489 usable items, 476 (97.3%) are unanimous and 13 (2.7%) are two-
to-one splits resolved by the majority; there are no three-way ties, so no
440 item required a tie-break rule or a further adjudication round. One item
was marked wrong by all three raters without a replacement label supplied
and is excluded rather than guessed. Disagreement is not discarded silently:
the per-item votes are released with the gold standard, so a reader can re-
compute any statistic on the unanimous subset alone. Before the main task
445 all three annotators completed a shared 20-review calibration round whose
disagreements were discussed and resolved against the written decision rules,
which is where the systematic conflicts (slow versus broken, device-specific
versus general) were settled rather than in the main pass.

A disclosure about an earlier annotation round. The 490-review
450 gold was annotated twice. In the first round three sheets were distributed,
and the two returned alongside the lead author’s carried annotation columns

identical to hers cell for cell, so no genuine agreement could be computed from them and they are not used anywhere in this paper. The round we report is the second, described below, and the agreement statistics come
 455 only from it. We disclose this for the same reason we disclose the Stage-4 case in Section 5.2: a paper that offers a check for non-independent ratings should say where its own artifact failed that check. Both rounds are released.

(ii) Independent human rater panel (three raters). The 490-review classification gold is annotated independently by the lead author and two
 460 further human raters on the identical item set. Each rater sees the classifier’s predicted label and marks it correct (Y) or wrong (N), supplying the correct class when N. All three complete a shared 20-review calibration round with adjudicated disagreements before the main task opens, and work from the same written decision rules (Appendix Appendix B). The gold label for
 465 each review is the majority vote over the three raters. This is a verification task rather than open labeling, which is why agreement is high relative to free-labeling studies of app reviews.

(iii) Stage 4 rater panel (same three people). Stage 4 was rated twice. The first round scored a 400-row set whose conditions were produced
 470 by template composers; we discarded it, for reasons given in Section 5.2. The round we report is a 200-row set in which both arms come from the same model and differ only in whether the IssueSpec was supplied. All three raters scored all 200 rows independently, blind to the condition behind each reply, and we release both rounds together with the independence check that
 475 separates them. Three human annotators contribute in total, the lead author and two collaborators; categories (iv) and (v) are model raters, not people.

(iv) LLM-as-judge (Qwen2.5-3B-Instruct). Scored 28 IssueSpecs on the five-dimension rubric and 50 sub-clusters on the Y/P/N purity rubric [46], both calibrated against the lead author.

(v) Corpus-labeling LLMs. The V2, cleanlab-V2, and V5 LLMs are
 480 treated as independent raters for inter-rater statistics.

Table 6 consolidates agreement and disagreement across all pairs.

Ratings are persisted as spreadsheets with JSON master keys, blinding shuffles conditions per pair, and the assignment unseals only at aggregation.
 485 Two distinct three-rater statistics appear in Table 6 and should not be conflated. The classification gold uses three independent *human* raters on 490 reviews and reaches Fleiss $\kappa = 0.968$ on the Y/N verdict and 0.979 on the effective seven-class label, with no three-way ties. The $\alpha = 0.285$ row is a separate check on 99 reviews that uses the lead author plus two Qwen2.5-

490 3B prompts (concise and chain-of-thought) as raters, measuring seven-class
classification agreement rather than cluster purity. We reported this row as
 $\alpha = 0.451$ and as a purity check; both were wrong. The value came from an
implementation that credited each rating with a coincidence against itself,
inflating observed agreement; the corrected figure agrees with the reference
495 `krippendorff` package to four decimals. The three Cohen’s κ in the same
table were computed with `scikit-learn` and are unaffected. The correction
moves the row from the moderate band into the fair band, which weakens
the LLM-rater panel rather than any claim we make.

4.3. Multi-LLM Bias Control

500 A single-LLM headline raises a structural-bias concern: results from any
one model may not transfer across families. We therefore replicate Stage 3
across four LLMs drawn from three distinct families, Anthropic Claude (Opus
4.7), Meta Llama-3.3-70B-Instruct (via Groq), and Alibaba Qwen2.5- $\{3B,$
 $1.5B\}$ -Instruct, with Stage 4 replicated on a Claude-vs-Qwen-3B subset, and
505 check that cross-family rank order is preserved (Table 9). Frontier-family
replication on GPT-4o and Gemini-1.5-Pro is queued.

4.4. Experimental Design

We pre-register three experiments, one per research question:

Experiment 1 (RQ1, translation quality). Five conditions are com-
510 pared on the same 100 stratified clusters: (a) LLM with taxonomy grounding
(the full Stage 3 system), (b) LLM without taxonomy grounding (free-form),
(c) raw review summary (a lower bound that concatenates the cluster’s
reviews), (d) lead-author reference, and (e) human-written GitHub issues
mined from three open-source Android projects (the practitioner-quality up-
515 per bound). Each spec is scored on the five-dimension rubric. Significance
is reported via paired Wilcoxon signed-rank tests on rubric scores between
conditions.

Experiment 2 (RQ2, coupled vs. uncoupled), as reported. The
design we report holds the generator fixed. The same model, given the same
520 retrieved context, writes two replies for each of 100 reviews, differing only in
whether the validated IssueSpec is in the prompt. All 200 rows are rated blind
by three raters on quality, specificity, and helpfulness, and analysed with a
paired Wilcoxon test and a paired bootstrap interval. *The discarded design.*
An earlier round used four conditions produced by template composers rather

525 than by a model; we describe it because the paper reports its withdrawal,
not because it supports a claim.

Experiment 2, discarded design. The same 100 clusters drive four response-generation conditions: (a) an rrgen-style baseline (review only), (b) a prompt-baseline (CoRe-style with dev-rel prompt and lightweight entity extraction), (c) RAG without IssueSpec, and (d) the full system with IssueSpec. Evaluation uses automatic metrics (BLEU, ROUGE-L, BERTScore) and 400 blinded paired human ratings (quality 1–5, specificity 1–5, helpful Y/N), with the lead author as primary annotator and a second rater on a 30-pair subset. Statistical significance is checked simultaneously by paired Wilcoxon, Friedman with Nemenyi post-hoc [52], Bradley-Terry, and McNemar at Bonferroni-corrected $\alpha = 0.05$. Effect sizes follow Romano et al. [53] and Cliff [54].

Why these four conditions. The Experiment 2 ladder is designed so each rung adds exactly one thing. The prompt baseline adds dev-rel system instructions on top of an rrgen-style review-only baseline, `no_spec` then adds retrieval, and `full` then adds the validated IssueSpec. Reading the ladder from bottom to top isolates which single addition moves human-rated quality. The headline comparison runs single-shot vanilla RAG [10] rather than an agentic loop, so that the IssueSpec’s contribution is not confounded with orchestration; the agentic-RAG comparison against Self-RAG [28] and DSP [29] runs separately on a 10-review feasibility sample (stratified, two per issue type, Qwen2.5-3B as critic and reviser, up to two iterations) and is reported as a demonstration rather than a powered comparison.

Experiment 3 (RQ3, dual- vs. single-objective RLHF). Five policies run on a distilGPT2 proof-of-concept (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, single-objective KTO [20], single-objective DPO [19], a constrained-proxy variant, and dual-objective Lagrangian Constrained PPO [21, 22]. We measure head-to-head BLEU, ROUGE-L, and BERTScore on a 100-review test set, plus the rule-based safety-violation rate.

5. Results

How to read this section. Each research question owns one subsection, one headline figure, and one headline table, and nothing else in the section is required to follow the argument. Section 5.1 answers RQ1, is the typed IR producible at scale, with Table 11 for template-fill and Figure 2 for the

classifier recovery that makes corpus-scale production possible. Section 5.2 answers RQ2, is the IR load-bearing downstream, with Table 14; the answer is no, and the subsection explains what our earlier positive answer was measuring. Section 5.3 answers RQ3, does constrained alignment demonstrate a quality-versus-compliance trade-off, and reports a negative result. Sections 5.4 and 5.5 are supporting evidence: they test whether individual components earn their place, and Section 6.1 states plainly which ones do. A reader tracking only the contribution can read Sections 5.1, 5.2, and 6.1.

Table 7 states what we take away from the section before the evidence for it, so a reader can decide which subsections to read.

5.1. *Is the IR Producible at Scale?*

Template-fill measures schema conformance, and its limit is worth stating exactly rather than in general terms. The criterion is the templates’ own required fields, and the free-form condition fills the four shared fields and none of the type-specific ones, so its score is determined entirely by the issue-type mix: $(30 \cdot 4/7 + 30 \cdot 4/6 + 40 \cdot 4/5)/100 = 0.691$, with zero variance across specs and a strict-minus-loose difference of exactly zero. The 0.69 is therefore a constant, not a measurement, and the 0.96-versus-0.69 gap says that routing produces type-specific fields at all. It carries no information about whether the routed text is better written. Under strict content-validity criteria (at least three reproduction steps with action verbs, descriptions ≥ 30 words, and a full As-a/I-want/So-that triple for user stories), the LLM-with-taxonomy condition achieves a substantive template-fill rate of 0.96 (Table 11). Free-form LLM reaches 0.69, the same as the lead-author reference set ($n = 20$). Raw concatenation lands at 0.34, and real GitHub issues mined from three open-source Android projects reach 0.53. The LLM-vs-GitHub gap reflects coverage rather than content quality: the LLM is prompted to fill every field, whereas humans leave optional fields blank. A very-strict sensitivity sweep preserves the ordering, Claude drops from 0.97 to 0.71, with the Claude-vs-GitHub gap shifting only -0.085 .

SPECCOV scores, and a correction to how they were computed. An earlier version of our scorer applied per-condition floors, raising `raw_summary` to 5 and the human reference to 4 regardless of content, on the reasoning that verbatim copying is faithful by construction and hand-written references deserve a floor. That is not a measurement, because a score whose value depends on the label of the thing being scored cannot compare labels, so we removed the floors and report what the scorer produces unaided.

Across 320 specs the LLM-with-taxonomy condition averages 4.19, free-form LLM 3.38, raw concatenation 4.47, and the human reference 3.45. The two LLM conditions are unchanged, because no floor was ever applied to them; the floors had inflated the two comparison conditions, so removing them makes our own system look better rather than worse, and we note that explicitly so the direction of the correction is on the record. Read on their own these numbers rank raw concatenation highest at 4.47, which is the trivial upper bound obtained by copying the source, with the taxonomy condition second at 4.19. Section 5.1 shows they should not be read on their own. Top-ranked aspects by PageRank centrality are **ad** (0.040), **phone** (0.011), and **battery** (0.007).

SpecCov does not measure faithfulness

An extractive-coverage score is only useful if it agrees with what a person calls faithful. We tested that directly. Three human raters independently scored 60 specs, 20 from each of the three generators, shuffled and unlabelled, on a 1–5 faithfulness scale, and separately listed every concrete claim the source reviews do not support. One qualification is needed before the numbers. One rater completed the corrected sheet, which shows exactly the evidence SPECCOV itself scores against; the other two completed an earlier draft that showed a subset of it, so they judged against less evidence than the metric uses and their flags are biased toward over-flagging. We therefore lead with the rater who saw the full evidence and report the other two beside her. The metric fails all three checks on either reading.

It does not rank specs the way humans do. Spearman ρ between SPECCOV and human faithfulness is +0.045 ($p = 0.73$), +0.015 ($p = 0.91$), and +0.226 ($p = 0.08$) for the three raters, and +0.147 ($p = 0.26$) against their mean. No correlation reaches significance.

It does not detect the claims humans call unsupported. On the 60 specs, one rater marked 45 as containing at least one unsupported detail and 15 as clean. SPECCOV scores those two groups at 3.64 and 3.93, a difference in the right direction but far from significant (Mann-Whitney $p = 0.34$). A metric that detected hallucination would separate these groups.

It orders the generators backwards. SPECCOV ranks the taxonomy condition (4.20 on this subsample) above the human reference (3.45); the humans rank them the other way, human reference 4.10 and taxonomy 3.50. The mechanism is visible in the definition: templated specs reuse source vocabulary densely, which is what the score rewards, while a human writer condenses

635 and paraphrases, which the score penalizes even when nothing is invented.

We therefore withdraw SPECCOV as a faithfulness measure. Lexical grounding is not a usable proxy for faithfulness on this task, and the automatic 320-spec rubric that appeared to support it is not independent evidence, since its faithfulness dimension is computed by the same overlap
640 procedure. Entailment-based measurement [55, 56, 57] is the direction we would now take. We release the scorer, the 60-spec human ratings, and this analysis, because a documented negative result on a plausible-looking metric is more useful to the next person than a metric that was never tested.

Cross-LLM replication (bias control). Table 9 reports four uses
645 across three model families. Llama-3.3-70B reaches 0.80 template-fill, Qwen2.5-3B 0.74, and Qwen2.5-1.5B 0.48 under the same strict criteria; the templated > free-form > raw ordering holds across all four models with capability-scaled magnitudes. On two per-class structural-compliance checks, Llama-3.3-70B attains complete template compliance on the strati-
650 fied sample: all 29 bug specs include three or more reproduction steps with action verbs (vs. Claude’s 73%), and all 30 feature specs include the As-a/I-want/So-that triple (vs. Claude’s 97%). These measure structural template compliance (does the field exist with the required form?), not content correctness, content quality is reported separately under the five-dimension
655 rubric. The structural-bias control that matters here is cross-family rank preservation: the IssueSpec contract transports across families rather than depending on a single proprietary generator.

Cross-family rubric quality. Template-fill measures whether a generator emits the required fields; it says nothing about whether the content
660 is any good. We therefore ran the same Qwen2.5-3B judge and the same five-dimension rubric over the other generators’ specs, on matched clusters (Table 8). On the 14 clusters all four generators cover, mean rubric score follows the same order as template-fill: Claude 4.09, Llama-3.3-70B 3.96, Qwen2.5-3B 3.83, Qwen2.5-1.5B 3.77. We state plainly what that ordering
665 is worth. At $n = 14$, or $n = 13$ for the three pairs involving Qwen2.5-1.5B, none of the six pairwise differences reaches significance (Wilcoxon p between 0.18 and 0.75), and the adjacent gaps are the same size as the run-to-run generation noise measured in Section 6.4. The four-way ordering is therefore consistent with the template-fill result rather than independent confirmation
670 of it. The Claude-versus-Llama comparison in Panel B, which has twice the sample and spans all five issue types, does reach significance at +0.38 ($p = 0.015$), and that is the one cross-family quality difference we assert. The

spread is much narrower than the template-fill spread (0.96 to 0.48), which is informative in itself: the capability gap between these models shows up
675 mostly as whether fields get filled at all, not as the quality of the text once they are. Completeness is the dimension that separates them (4.00 down to 3.00), while accuracy is nearly flat (4.50 for three of the four). On the wider Claude-versus-Llama panel, which spans all five issue types, Claude leads 4.01 to 3.64, with the gap concentrated in actionability and completeness.
680 Two caveats belong with these numbers. The four-way panel is bug-report clusters only, because that is where the two Qwen runs overlap, and the judge is itself Qwen2.5-3B-Instruct, so the Qwen2.5-3B row carries a self-preference risk that the others do not.

Five-Dimension Rubric. The proposal specifies a 5-dimension Likert rubric (completeness, accuracy, actionability, specificity, clarity, each 1–
685 5) [44, 45, 46], operationalized with Qwen2.5-3B-Instruct as the judge. We score the complete set rather than a subsample: all 100 taxonomy specs and all 20 lead-author reference specs, in a single run (Table 10). The taxonomy condition reaches 3.94/5 (sd 0.59 across specs), above the hypothesized lower
690 bound of 3.5. An earlier stratified 28-spec subsample under the same judge and rubric gave 3.89, so subsampling cost 0.05 of a Likert point here, which bounds how much the small-sample version of this number could have been misleading. Per issue type the taxonomy condition ranges from 4.23 on performance and 4.02 on bug reports down to 3.45 on usability, the type whose
695 reviews are least specific about what went wrong.

The reference set is a judge calibration, not a competitor. It scores 3.45 overall, below the generated specs, and the gap is concentrated in actionability (2.85 versus 3.86). The reason is visible in the sheets: the lead author filled title, description, severity, and component but left the template-specific
700 fields such as reproduction steps and acceptance criteria empty, because writing a reference spec by hand is a different task from filling a routed template. The comparison therefore says that a human writing freely produces less tracker-ready output than a type-routed generator, which is the same effect the 0.96-versus-0.53 template-fill gap measures against real GitHub issues. It
705 is not evidence that the specs are better than expert judgement on content, and we do not read it that way.

Stage 1 Classification Recovery. The Cohen’s κ trajectory against the 490-review expert gold crosses three Landis-Koch [51] agreement bands (Table 12): the V2 LLM achieves $\kappa = 0.163$ (slight); after cleanlab correction [23] $\kappa = 0.333$ (fair); the V5 classifier retrained on the corrected
710

labels reaches $\kappa = 0.592$ (moderate, just below the substantial threshold). The trajectory does not depend on any single annotator. Scored against the majority-vote gold of the three-rater human panel ($n = 489$ after excluding one review left without a correction label), the same three classifiers reach
715 $\kappa = 0.165$, 0.334 , and 0.590 , within 0.002 of the lead-author values at every step (Table 12). On the strictly held-out 307-review subset (183 of the 490 entered training via the verified anchor), V5 reaches $\kappa = 0.616$, slightly above the 0.592 on the full 490, ruling out a contamination artifact.

The per-class recovery story (Table 13) makes the structural collapse
720 visible. On the two minority classes the V2 LLM cannot see at all, F1 jumps to 0.83 for **compatibility** and 0.77 for **performance**. Applied across the full 215,583-review corpus, V5 agrees with the cleanlab correction on 88.66% of the 40,291 corrected cases. We want to be exact about what that number is and is not, because it is easy to over-read. V5 was trained on a corpus
725 in which 38,985 of those corrected rows carry the corrected label, so its agreement is substantially a measure of fit to its own training signal, not an independent second opinion. We report it as a consistency check, which is what it is, and we do not claim the two signals are independent.

The stronger evidence is the held-out gold, and we describe the split
730 precisely because the mechanism matters. V5’s training corpus is built with a stratified per-class cap of 15,000 rows at seed 42. Replaying that cap identifies which corpus rows survived into training: 183 of the 490 gold reviews did, and 307 were capped out and never seen, of which 306 carry a gold label and are scorable. The split is therefore an artifact of class-balancing rather
735 than of anchor membership. It is not, however, unbiased in the way we first described. The cap binds only on the four classes V5 predicts most often, so membership of the unseen set is decided by V5’s *predicted* class rather than at random. Scored by expert label, which is what κ is computed against, the unseen 307 still spans all seven classes and includes 38 **compatibility**, 32
740 **usability**, and 31 **performance** reviews, so the held-out comparison does cover the classes carrying the recovery. What it cannot rule out is a subtler selection effect: the reviews V5 predicted confidently into large classes are over-represented among the unseen, and those may be easier. On the unseen reviews V5 reaches $\kappa = 0.616$, computed on the 306 of them that carry a gold
745 label, slightly *above* the 0.592 it scores on the full 490. A model that had merely memorised its training labels would fall on unseen items rather than hold. One caveat belongs with this. The split is reconstructed by replaying the cap rather than read from a saved training manifest, so it is a faithful

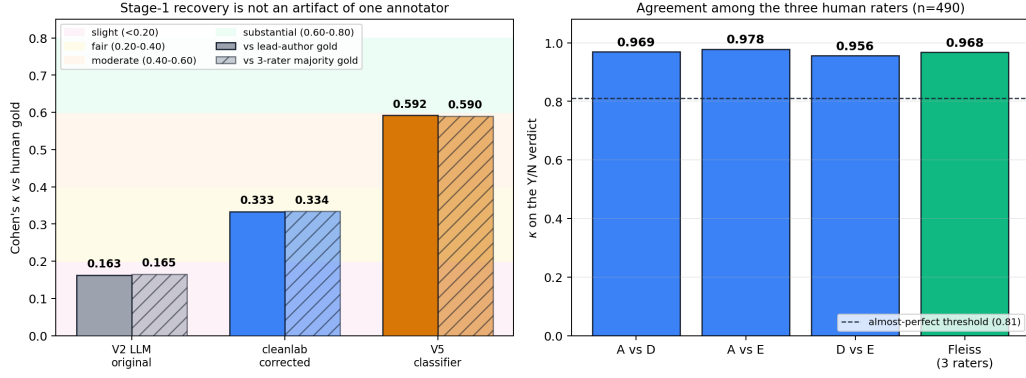


Figure 2: Stage-1 classifier recovery and the gold standard it is measured against. *Left*: Cohen’s κ for the three classifier versions, plotted twice, once against the lead-author gold (solid) and once against the three-rater majority-vote gold (hatched). The two are within 0.002 at every step, so the 0.163 to 0.592 trajectory does not depend on any single annotator. Background bands are the Landis-Koch [51] agreement ranges. *Right*: agreement among the three human raters on the same 490 reviews, pairwise and as a three-rater Fleiss κ , all above the almost-perfect threshold. Read the left panel for the result and the right panel for the reliability of the yardstick it uses.

reconstruction rather than a recorded fact; we release the replay script so the
 750 reconstruction can be checked. The choice of confident-learning thresholds
 is not load-bearing: a full 3×3 grid over $(\text{anchor_conf}, \text{max_llm_prob}) \in$
 $\{0.65, 0.70, 0.75\} \times \{0.15, 0.20, 0.25\}$ keeps V5 endorsement in an 82.7–91.0%
 band, with the headline (0.70, 0.20) at 88.66% sitting near the grid center.

To rule out memorization of the 200 synthetic compatibility templates
 755 used during V5 training, we evaluate V5 against Maalej’s 5,008 labels under
 a label-space cross-protocol test. Review texts appear in the anchor
 under the RRGGen seven-class taxonomy, but predictions are scored against
 Maalej’s separate taxonomy, which contains no `compatibility` class, making
 any compatibility predictions out-of-distribution with respect to the evaluation
 760 labels. On the five overlapping classes ($n = 4,560$), V5 achieves macro
 $F1 = 0.676$, weighted $F1 = 0.730$, and accuracy = 72.0%. V5 also flags 432
 reviews (8.6%) as `compatibility`; the top-twenty-confidence flags read as
 canonical compatibility complaints (“crashes on iphone 5 and ipad 3”, “works
 on HTC Desire X but not Samsung Galaxy Tab 3”), all labeled `bug_report` in
 765 Maalej because Maalej offers no alternative class. The detector is performing
 concept-recognition, not template-matching.

5.2. Is the IR Load-Bearing? A Result We Had to Withdraw

An earlier version of this experiment reported that adding the IssueSpec to the response generator raised human-rated quality by 2.35 Likert points. That number was wrong, not in its arithmetic but in what it measured, and the correction is the most important thing this section has to report.

What was wrong. The four conditions in that experiment were not produced by a language model. Each was produced by a deterministic template composer: a Python script that selects sentence fragments from a hand-written bank and fills slots, with no language model in the loop. The composers differed in authorial register as much as in what information they were given. The `no_spec` arm emitted lemmatised, ungrammatical text averaging 78 words (“we be sorry that the login loop you describe have got in the way”); the `full` arm emitted polished prose averaging 123 words, with cleanup passes and an empathy phrasebook that no other arm received. A rater comparing them was separating two writing styles, and the presence or absence of the IssueSpec was confounded with that separation. The gap measured the composer at least as much as the specification.

The clean comparison, and what we can and cannot verify about it. We re-ran the experiment with the generator held fixed. Both arms come from the same model on the same 100 reviews, and the intended difference is whether the validated IssueSpec appears in the prompt. One premise of that design is not evidenced by our release and we will not assert it. Neither response file stores the prompt or the retrieved text, and the two files record their retrieval sources under different label schemes with different counts, so on the released artifact no row has an identical recorded source list. Both arms draw from the same candidate pool of past and semantically similar responses, and that much is checkable, but whether the retrieved context was identical per review is not recoverable. A reader should therefore treat the comparison as holding the generator and the review set fixed, not as a strict single-variable manipulation. Retrieval differences would add noise rather than a systematic advantage to either arm, which makes a null easier to obtain, so this does not rescue the withdrawn positive result; it does mean the null is weaker evidence than a fully controlled design would give. The two arms average 105 and 97 words with distinct-1, the ratio of unique unigrams to total unigrams, at 0.736 and 0.760. They are therefore close on the surface features a rater could use to tell them apart, though the length difference is itself significant (paired Wilcoxon $p = 6 \times 10^{-8}$), which we return to when

discussing what the lead author’s preference tracks. Three raters scored all
805 200 blinded rows.

The IssueSpec does not measurably improve response quality.
Pooled over the three raters the gain is +0.04 Likert points, 95% bootstrap CI
[−0.10, +0.18], paired Wilcoxon $p = 0.38$ on 100 pairs, Cliff’s $\delta = 0.04$. Each
review’s two replies are shown in a randomised order, and we call the two
810 display positions slots. The assignment came out 64/36 rather than balanced
and the raters show a small preference for the first slot, so we also report
a slot-stratified estimate, which moves the pooled figure to +0.03 and each
rater’s by at most 0.03. We use the unadjusted numbers throughout so that
every figure in Table 14 is on one estimator, and release the stratified values
815 with the data. Specificity behaves the same way, pooled −0.00 ($p = 0.87$),
and the helpful rate is flat at 94% in both arms. On the evidence of this
design we cannot claim that supplying the typed specification improves the
response a user receives. We should be equally careful in the other direction.
With 100 pairs and the observed spread, the smallest effect this design could
820 detect at 80% power is about a fifth of a Likert point, so what we have ruled
out is a large effect, not a small one. Detecting a gain the size of the one we
measured would need on the order of 2,500 pairs. The honest statement is
that the effect, if any, is small enough that this study cannot see it, not that
it is zero.

**The raters disagree with each other, and that is the interesting
part.** Table 14 breaks the result out. The lead author, who designed the
system, prefers the IssueSpec arm decisively: +0.56 quality ($p = 2 \times 10^{-11}$, 59
wins, 5 losses, 36 ties) and +0.39 specificity ($p < 10^{-7}$). The two independent
raters lean the other way, −0.29 and −0.15, neither significant at $\alpha = 0.05$
830 ($p = 0.06$ and 0.09). Inter-rater reliability is correspondingly low, quadratic-
weighted κ of 0.03, 0.04 and 0.50, against 0.97 in the discarded template-
based round. High agreement there was a symptom of the confound: when
one arm is visibly broken, raters agree easily. Here they do not.

We looked for a content mechanism behind the lead author’s preference
835 and did not find one. Correlating each rater’s per-review preference against
the spec-derived features that the IssueSpec should introduce, named de-
vices, reproduction steps, and component mentions, produces no correlation
that survives inspection for any rater; the largest is $\rho = 0.19$ ($p = 0.06$)
between the lead author’s preference and reply length. So we can describe
840 the disagreement but not explain it. Two readings remain open. The lead
author may be perceiving triage value that a general reader has no reason to

weigh, in which case the right evaluation asks developers rather than readers. Or the preference may be expectancy on the part of the person who built the system. This design cannot separate those, and we do not assert either.

845 **Why we believe the raters this time, and what the check does and does not show.** In the discarded round the two additional raters deviated from the lead author on only 16 and 19 of 400 rows, those deviation sets did not overlap on a single row, and every deviation was exactly one point. We initially read the zero overlap as the damning fact. It is not:
 850 with deviations that rare, the hypergeometric probability of zero overlap under perfect independence is 0.45, so zero overlap is the single most likely outcome and carries almost no information. The discriminating observation is the second one. On the quality column every deviation in both sheets had the same magnitude, one point, which is what a vector perturbed by a
 855 fixed step produces and not what human disagreement produces. We state the column because the specificity column in the same sheets does show magnitudes of one, two and three, so the signature is present in one column and not the other. In the reported round the raters differ from the lead author on 133 and 145 of 200 rows, the deviation sets overlap on 101 where
 860 chance predicts about 96, and the deviations take four distinct magnitudes from one to four points. We release the check with the data, and we state its weak prong plainly, because a check offered as a contribution should come with its own limits attached.

What survives. Less than we first thought. The claim that the typed IR
 865 carries downstream response quality does not survive, and neither does the retrieval finding we had treated as the surviving half: it was measured against the prompt baseline rather than a no-context arm, and against the no-context arm it is -0.07 at $p = 0.90$. Both sat in the discarded round. Stage 4 leaves us with a null and nothing else. Stage 3’s contribution, producing tracker-ready
 870 structured records at corpus scale, stands on its own evidence in Section 5.1; it no longer has a demonstrated downstream payoff in response generation.

5.3. Aligning the IR-Aware Generator at Scale

Five policies were trained on distilGPT2 (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, KTO [20],
 875 DPO [19], a constrained-proxy variant, and Lagrangian Constrained PPO [21, 22]. Head-to-head on a 100-review test set, per-policy BLEU-1 / ROUGE-L / BERTScore F1 are: SFT-base 0.090 / 0.097 / 0.802;

KTO 0.068 / 0.074 / 0.792; DPO 0.084 / 0.088 / 0.800; **constrained-proxy 0.137 / 0.131 / 0.806**; Lagrangian PPO 0.087 / 0.091 / 0.798.
 880 The constrained proxy leads three of four metrics (BLEU-1, ROUGE-L, BERTScore F1; the rule-based safety-violation rate is uninformative here because compliance is satisfied at initialization), with +53% BLEU-1 over SFT-base. KTO underperforms here, consistent with its small-data assumption breaking on a 296-pair training set.

885 **The dual-objective claim is not demonstrated, and we report that as the result.** At distilGPT2 scale the compliance scorer is satisfied at initialization ($C = 0.94 \geq \tau$), so the Lagrange multiplier λ collapses to zero and the update reduces to single-objective optimization. Two further experiments confirm this is a property of the regime rather than of one
 890 threshold choice. A re-run that replaces the permissive scorer with the four operational compliance checks and tightens τ from 0.50 to 0.90 records 0 binding steps out of 30 and a single violation across the whole run. A sweep over $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ leaves the pass rate at 100% for four of the five policies at every threshold, with the first binding τ undefined for those
 895 four; only SFT-base ever dips, to 99% at $\tau \geq 0.9$. A model of this size cannot plausibly violate the rubric, so the constraint has no work to do and the CMDP machinery is untested under binding conditions. The +53% BLEU-1 figure for the constrained proxy is therefore a single-objective result, not evidence of a quality-versus-compliance trade-off, and we do not claim it as
 900 one.

The constraint binds once the policy can violate it. The obstacle is not parameter count, it is capability: distilGPT2 never produces the fluent developer-relations phrasing the rubric penalises, so it cannot violate and the constraint has nothing to hold back. We measured this directly. We first
 905 justified this by observing that 5.2% of our own Stage-4 generations trip the scorer. That figure is from the four withdrawn template-composed arms (21 violations in 400 replies). Scored the same way, the two model-generated arms we actually defend violate on 4 of 200 replies, 2.0%, and one of the four is an information-leak rather than an over-promise. The zero-shot Qwen2.5-
 910 1.5B probe gives 2 violations in 40 replies, 5.0%, whose 95% interval runs from roughly 1% to 17% and therefore cannot be distinguished from 2.0% either way. distilGPT2 gives roughly zero. The honest version of the argument is weaker than the one we first made: a 1.5B instruct model violates the rubric occasionally where distilGPT2 essentially never does, and that is enough for
 915 the constraint to bind, but we cannot claim it matches the rate of our own

generations. We therefore re-ran the Lagrangian loop with Qwen2.5-1.5B-Instruct as the policy, LoRA adapters over the attention projections, the frozen base as the KL reference, and a zero-tolerance threshold $\tau = 1.0$; the scorers, the dual-ascent rule, and the batch size are unchanged from the distilGPT2 run. The behaviour inverts. Over 90 steps the constraint binds on 13 of them (14.4%), and λ leaves zero, rises monotonically to 0.65, and never collapses back. A 30-step run under the same setup reaches $\lambda = 0.125$, so the multiplier scales with the horizon rather than saturating. Split by thirds, the trajectory shows the dual ascent responding to violations as they arrive: 1 binding step and $\lambda = 0.05$ in the first thirty, 7 and $\lambda = 0.40$ in the second, 5 and $\lambda = 0.65$ in the third. The qualitative claim that the CMDP degenerates to single-objective optimization is therefore a property of the proof-of-concept policy, not of the formulation.

What is and is not shown. The two series move in opposite directions across the run, quality from 0.565 to 0.590 and compliance from 0.997 to 0.989, while λ climbs. We are careful not to call this a trade-off. The step-level correlation between quality and compliance is $\rho = -0.12$ ($p = 0.25$), so what we have is two separate trends over a single run and no measured coupling between them, which is why the abstract reports the trade-off as undemonstrated. What we cannot yet show is the other half of the story, the multiplier growing large enough to pull the policy back into compliance. At a 5% base violation rate and a batch of four, roughly one violating sample arrives every seven steps, and 90 steps of LoRA updates at this learning rate do not accumulate enough constraint pressure to reverse the drift. So the correct statement is narrower than a confirmed RQ3 and wider than the earlier null: the dual objective engages, the multiplier responds monotonically to violations, and the tension between the objectives is measurable, while closing the loop back to compliance needs a longer horizon or a higher base violation rate than a 1.5B instruct model supplies.

What this layer contributes is the CMDP formulation instantiated over four domain-specific dev-rel constraints, an end-to-end trainer that runs, the threshold sweep that establishes the non-binding regime, the capability probe that explains it, the binding-constraint re-run, and the released testbed.

5.4. Cluster Quality and Purity

Stage 2 was redesigned from a flat UMAP+HDBSCAN baseline (194 clusters, mean size 471.0) to the three-layer KG-hierarchical design (605 sub-clusters, mean size 15.9). The two were not run over the same reviews, and

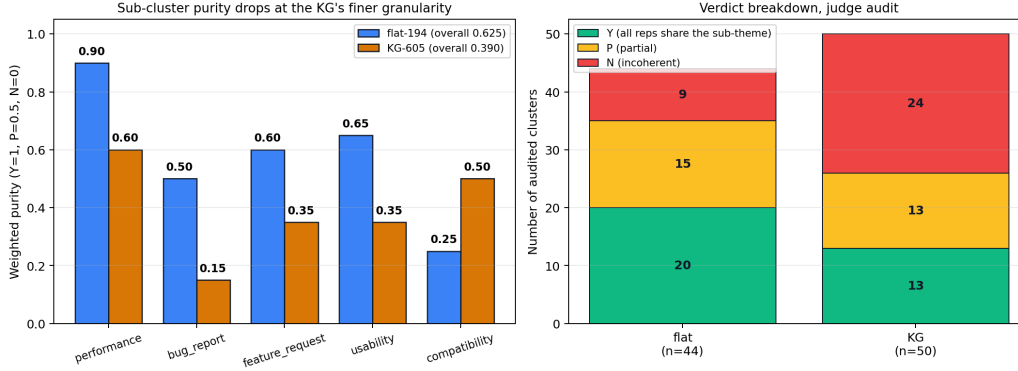


Figure 3: Extrinsic cluster purity for the flat baseline and the KG hierarchy, from the saved judge audit. *Left*: weighted purity per issue type, flat-194 against KG-605. The KG loses on four of five types, which is the finer-granularity cost we report rather than a result in its favour. *Right*: the underlying Y/P/N verdict counts behind the two overall numbers, 0.625 for flat and 0.390 for the KG. This figure is the evidence for the A2 ablation and for the claim in Section 6.1 that the KG’s contribution is prioritization, not cluster quality.

we state that plainly because it bears on how the table below should be read: the flat baseline clusters 91,368 reviews (117,958 in, 26,590 assigned to HDBSCAN noise) drawn from the full relabelled corpus, whereas the KG-hierarchical design is built on the 8,404-review knowledge-graph sample, of which 5,689 distinct reviews land in a sub-cluster. A sixteen-fold difference in the underlying review set is a third reason, on top of the two given below, why the head-to-head intrinsic comparison is not a comparison. We compute two complementary validity axes. *Intrinsic geometry*, Davies-Bouldin [58], Calinski-Harabasz [59], silhouette [60], follows standard IR/clustering practice [61, 62, 63]. *Extrinsic cluster purity* [64, 61] uses a Y/P/N weighted-purity rubric, $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$, with Y = all 5 share theme, P = 3–4 of 5 share, N = fewer than 3 share, following inter-rater conventions compatible with Landis-Koch [51].

We reported that KG-hierarchical achieves $5.4\times$ lower Davies-Bouldin and $1.9\times$ higher Calinski-Harabasz than the flat baseline. We withdraw those two figures. The script that produced the flat baseline’s intrinsic metrics draws its cluster labels as a random subsample of 10,000 reviews spanning all 194 clusters, but collects the review texts to embed by walking the clusters in order, and the first flat cluster alone contains 10,507 reviews. The embeddings therefore come from a single cluster while the labels span 194, so the flat column’s DB of 12.15 and CH of 0.98 measure random labelling

rather than the flat partition. The KG column was computed on a correctly
975 paired set, so the comparison was a valid measurement against a broken one.
Table 15 keeps the KG column and marks the flat intrinsic column as with-
drawn. Repairing the alignment does not make the two columns comparable
in any case: the flat side embeds 10,000 full reviews while the KG side em-
beds three representative reviews per cluster, so the two are computed over
980 different kinds of object. That asymmetry is a second reason not to reinstate
the comparison, and the count-matched A1b ablation, which embeds rep-
resentatives on both sides, is the only intrinsic comparison we report. The
count-controlled A1b ablation below is unaffected, uses a separate correctly
paired script, and is what our Stage-2 conclusion now rests on. Extrinsic
985 purity is also unaffected: the KG’s finer clusters score 0.39 against the flat
baseline’s 0.625 under the same Qwen judge. By Landis-Koch [51] convention
adapted to weighted purity, the flat baseline’s 0.66 lead-author purity sits in
the substantial-agreement band (≥ 0.61) and KG-605’s 0.39 in the fair band
(≥ 0.21), reflecting the granularity-vs-coherence trade-off the A1b ablation
990 makes explicit.

Count-controlled diff from original. A count-controlled ablation
(A1b) disambiguates two evaluation axes that prior cluster-quality work
tends to conflate: intrinsic geometry and practitioner-side prioritization
(centrality-weighted coverage). On 1,815 representative reviews (3 per
995 sub-cluster across 605 sub-clusters), an agglomerative clustering forced to
605 clusters reaches $DB = 1.12$, $CH = 4.21$, $silhouette = +0.148$, against
KG-605’s $2.24 / 1.85 / -0.234$. This is now the only intrinsic-geometry
comparison we make, since the flat-194 figures are withdrawn above: on
geometry, count-matched agglomerative clustering beats the KG on all three
1000 metrics.

Where the KG actually wins. On the prioritization axis the story re-
verses. Of the 10,534 aspect nodes in the KG, the top five by PageRank are
touched by 55% of reviews and the top ten by 66%. We give the deduplicated
figures; counting a review once per aspect it mentions inflates these to 74%
1005 and 96%, and that inflated pair should not be used. A developer triaging
in PageRank order completes three-quarters of the complaint volume after
reading five aspect buckets, whereas random aspect ordering would require
thousands of buckets to reach the same coverage, a roughly $10^3\times$ concentra-
tion ratio. The KG additionally enables per-aspect drill-down, distinguishing
1010 **battery**→Samsung-drain from **battery**→fast-charge.

5.5. Ablations

An ablation removes one component and re-measures, so that a component’s contribution is demonstrated rather than assumed. Eight are listed below, counting A1b, which a reviewer asked for. Seven were run and one, A7, is deferred for a reason given at the end. Each one below states what it removes, what the removal is supposed to test, what happened, and what we conclude, because three of the seven came back against the design and a summary line would hide that. Table 16 collects the verdicts.

A1, remove the knowledge graph and feed flat clusters into Stage 3. This asks whether Stage 3 reads the graph or merely the group. Stage 3 quality does not change, because the translator consumes a cluster centroid and its representative reviews, not the graph edges. The KG is therefore not a Stage 3 dependency, which narrows where its value could lie.

A1b, compare the KG against flat clustering matched to the same number of clusters. This is the reviewer-prompted version of A1 and it exists because of a confound. The KG produces 605 sub-clusters and the flat baseline produces 194, and intrinsic metrics move with cluster count on their own, so any comparison between the two conflates structure with granularity. Re-running agglomerative clustering forced to exactly 605 clusters removes the confound. The flat version then wins on all three metrics, Davies-Bouldin 1.12 against 2.24 (lower is better), Calinski-Harabasz 4.21 against 1.85 (higher is better), and silhouette +0.148 against −0.234 (higher is better, and a negative value means the average point sits closer to a neighbouring cluster than to its own). The honest conclusion is that the KG does not produce geometrically better clusters. What it produces is an ordering, through PageRank centrality over the review-aspect graph, and prioritization is the only cluster-layer claim we make.

A2, remove hierarchical sub-clustering. Weighted Y/P/N purity, judged on five representative reviews per cluster, is 0.625 for the flat 194 and 0.390 for the KG’s 605. Finer granularity costs coherence here rather than buying it. Taken with A1b this is the second result against the cluster layer, and we report both.

A3, remove taxonomy grounding. The LLM writes a free-form issue description instead of filling a routed template. Strict template-fill falls from 0.96 to 0.69, and the per-type structural columns fall to exactly zero, because an unrouted generator never emits a reproduction-step list or an As-a/I-want/So-that triple at all when nothing asks it to. This is the ablation that

isolates the paper’s central mechanism. The gain is not the language model, it is the routing.

1050 **A4, remove the HITL-2 spec review.** Benchmark scores do not move, and the reason is procedural rather than substantive. HITL-2 is a production-time gate; the 100 specs used in the evaluation were scored before any human correction, so the headline numbers already describe pre-HITL output. The ablation therefore says nothing about whether HITL-2 helps in deployment,
1055 and we do not claim that it does.

A5, remove retrieval and keep the IssueSpec. BLEU-1 moves by -0.006 , ROUGE-L by -0.008 , and BERTScore F1 by -0.004 , all under one percent absolute. Three caveats belong with this result. The first is that A5 was generated by the template composer that Section 5.2 withdraws,
1060 not by the model-generated pipeline, so it measures retrieval’s effect on a generator we no longer report. First, A5 was scored on automatic metrics only, while A6 was scored by humans, so the pair is not measured on one instrument and the contrast between them is directional rather than a like-for-like comparison. A human-rated no-retrieval condition is the missing
1065 cell. Second, we withdrew the companion claim that retrieval without the IR scores below a no-context baseline. That comparison used the prompt baseline, which is not a no-context condition, and against the arm that is, the difference is -0.07 at $p = 0.90$. So A5’s reading is simply that retrieval makes no measurable difference here, with no accompanying story about it
1070 doing harm.

A6, remove the IssueSpec and keep retrieval. This ablation is superseded. Under the confounded design it showed quality falling from 4.59 to 2.24; under the corrected same-model design (Section 5.2) removing the IssueSpec changes quality by -0.04 , which is not distinguishable from zero.
1075 A5 and A6 together now say something different from what we first concluded: neither retrieval nor the typed intermediate produces a measurable change in the reply a user receives, and the Stage 3 to Stage 4 coupling we claimed is not supported.

A7, single-stream RLHF feedback. Not run. It was designed to test
1080 whether quality and compliance feedback can share one channel, which only becomes a meaningful question once the constraint binds and the two signals actually compete. Section 5.3 reaches the first half of that condition and not the second, so A7 stays deferred rather than reported.

6. Discussion

1085 6.1. What Carries the Result, and What Does Not

A five-stage pipeline invites the question of whether every stage earns its place. Our own ablations say it does not, and separating the two groups is more useful to a reader than defending the whole.

1090 *Load-bearing.* Two components carry the measured gains. Type-routed templating is the principal driver of Stage 3 quality: removing taxonomy grounding (A3) drops strict template-fill from 0.96 to 0.69, and the structural per-field metrics collapse to zero because an unrouted generator does not emit the reproduction-step list or the As-a/I-want/So-that triple at all. By contrast the typed IR is *not* a demonstrated driver of Stage 4 quality: 1095 under a same-model comparison removing it moves quality by 0.04 points ($p = 0.38$, Section 5.2), and removing retrieval instead (A5) moves every automatic metric by less than one point absolute. Our earlier reading of the retrieval arm does not survive either, and the error was ours to make twice. We first reported that retrieval alone was indistinguishable from a no- 1100 context baseline, then corrected that to significantly worse. Both statements compared the retrieval arm against the *prompt* baseline, which is not a no-context condition; it carries a developer-relations instruction. Against the actual no-context arm, which sees the review text only, retrieval scores 2.24 against 2.31, a difference of -0.07 at $p = 0.90$. And all three arms come from 1105 the discarded template-composer round, so none of them supports a claim in any direction. Nothing about retrieval survives from Stage 4 beyond the null.

Not load-bearing, and reported as such. The knowledge graph does not win on intrinsic cluster geometry. Under count matching (A1b), flat agglomerative clustering beats the KG on all three intrinsic metrics, and sub-cluster 1110 purity is lower at the KG’s finer granularity (A2). The KG’s defensible contribution is centrality-based prioritization, that is, which issues a developer should see first, not tighter clusters; we make no cluster-quality claim for it. We should be straight about how far even that goes. PageRank runs 1115 on the bipartite review-aspect graph, and nothing in our results shows it needs the hierarchical sub-clustering layer to do so. A flat clustering with the same aspect graph would plausibly support the same ranking, and we did not test that configuration. Stage 2 as built is therefore justified in its prioritization role and unjustified in its sub-clustering role, and a reader designing a lighter system should read it that way. Stage 5 sits in between. As 1120

Section 5.3 reports, the compliance constraint does bind once the policy is capable of violating it, and the dual update engages rather than collapsing, but no quality-versus-compliance trade-off is demonstrated at the scale we can train at. It is released as a testbed with a partial result. SPECCOV falls in the same group and for a sharper reason: tested against human faithfulness judgements it shows no correlation at all (Section 5.1), so we withdraw it as a measure rather than defend it.

What this means for the framework claim. The contribution that survives ablation is the typed IR and the type-routed templating that produces it, together with the verified-anchor procedure that makes the IR producible from noisy labels at corpus scale. Stages 2 and 5 are supporting infrastructure with honestly mixed evidence. A reader who wants the smallest system that reproduces the headline result needs Stage 1, Stage 3, and a generator conditioned on the resulting specification. That is less than Stage 4 as we built it: A5 shows the retrieval layer inside Stage 4 contributes under one percent on automatic metrics, so the part of Stage 4 that earns its place is the conditioning, not the retrieval.

6.2. Implications

Three claims travel beyond app reviews. *First*, LLM annotation noise on software-engineering text is not uniform, but the shape is not the one we first reported. We had described the 25% mislabelling rate as concentrated on minority classes. Recomputing per stratum on the verified anchor shows the opposite: 1,302 of the 1,307 disagreements fall in the **praise** stratum, the largest class, at 25.8%, against 0.5% on **performance**. What is true, and narrower, is that the labeller never emits two of its seven classes at all across the corpus, 8 **compatibility** and 184 **performance** labels in 215,583 reviews. Systematic absence of a class, rather than elevated error on rare classes, is what a small verified anchor is able to repair. *Second*, a caution rather than a claim. We expected a typed structural intermediate to be what does the load-bearing work in spec-aware response generation, and reported it as such. Section 5.2 withdraws that. The only downstream movement we can still point to is +0.12 on a rule-based scorer from agentic critique-revise iteration, on ten reviews, which is too small a sample to build on. What generalises is the negative lesson: a structured intermediate can be demonstrably better as an artifact and still leave the generated text indistinguishable, and pipelines of this shape should measure the two separately.

Third, the count-controlled cluster-validation ablation (A1b) delivers a self-contained methodological contribution: whenever a new clustering design changes cluster count and structure simultaneously, intrinsic geometry metrics (DB, CH, silhouette) conflate the two effects. The minimum reporting standard our analysis suggests is to (a) compare against a count-matched baseline on the same intrinsic metrics, and (b) pair that with a domain-utility metric (PageRank coverage in our case). This pattern applies directly to clustering-design comparisons in IR, software engineering, and NLP.

6.3. Threats to Validity

Construct. Template-fill measures coverage rather than content correctness, and Section 5.1 shows that SPEC-COV does not fill that gap: it is a lexical-grounding proxy that does not track human faithfulness judgement [47, 48, 49, 50]. The paper therefore has no validated content-correctness measure for Stage 3, and the five-dimension rubric carries that burden alone. NLI- and QA-based faithfulness measurement [55, 56, 57] is the clear next step. *Internal.* The count-versus-structure confound is exposed and resolved by the A1b ablation, which reframes the knowledge graph’s contribution as centrality-based prioritization rather than intrinsic geometry. A second internal threat we did not remove is that the 100 clusters carrying the Stage 3 and Stage 4 results were produced by the KG pipeline, the same layer A1b and A2 find wanting. Since flat clustering scores higher on purity (Figure 3), the headline gains may be measured on a slightly worse partition than a simpler system would produce, which would understate rather than inflate them. Re-running Stages 3 and 4 over flat clusters would settle it and is the cheapest open experiment in the paper. *External.* RRGGen contains 58 anonymized Android apps; V5 transfers zero-shot to the Maalej label space. Stage 2–4 cross-corpus transfer remains to be validated. The headline LLM is Claude Opus, with Llama-3.3-70B and Qwen2.5-{3B, 1.5B} as cross-family bias controls; GPT-4o and Gemini-1.5-Pro replication is queued (Section 6.5). *Conclusion.* The three-rater agreement on the classification gold ($\kappa = 0.968$ on the Y/N verdict) sits well above what free-labeling studies of app reviews report, which invites the question of whether the raters were genuinely independent. Three properties of the design account for the gap. The raters verify a supplied label rather than choose one from seven, which collapses most of the decision space; a 20-review calibration round with adjudicated disagreements precedes the main task; and the written protocol fixes eight

boundary rules that carry the cases annotators most often split on. Residual
1195 disagreement is rater-specific rather than random, one rater accounts for
most of the **usability** reassignments, which is the pattern independent
annotation produces. Stage 4 sits differently. All three raters scored the
complete 200-row re-run, and the gain does *not* reproduce per rater: the lead
author measures +0.56 and the two independent raters -0.29 and -0.15 .
1200 That disagreement, not the pooled null, is the honest summary of what
we know about Stage 4. *Reliability, and which α belongs to what.* Three
agreement statistics appear in this paper and they measure different things,
so a single pooled reading of them would be wrong. The classification gold
is annotated by three humans and reaches Fleiss $\kappa = 0.968$ on the verdict
1205 and 0.979 on the seven-class label. The Stage-4 response set that we report
is rated by the same three humans on all 200 rows, with quality weighted κ
of 0.03, 0.04 and 0.50. That is low, and it is the number to quote: the 0.970
to 0.986 range belongs to the discarded round, where one arm was visibly
broken and agreement was easy. The weakest figure, $\alpha = 0.285$ (corrected
1210 from a mis-implemented 0.451), comes from a deliberately weaker panel: a
seven-class classification check whose three raters are the lead author plus
two Qwen2.5-3B prompts, following the LLM-as-rater convention of Gilardi
et al. [44]. It bounds how far an LLM rater tracks a human on a subjective
sub-theme judgement; it is not the reliability of any headline result, and
1215 no claim in this paper rests on it alone. *Statistical.* Experiment 2 now
reports a null, so the relevant question is power rather than correction for
multiplicity. With 100 paired reviews and the observed per-review spread,
the 95% bootstrap interval $[-0.10, +0.18]$ excludes effects larger than about
a fifth of a Likert point in either direction. We can therefore rule out a large
1220 effect, not a small one; an effect of a tenth of a point would need several
times this sample.

6.4. Reproducibility and Extensibility

Every numerical claim in Section 5 is backed by a saved data file, and the
full path from raw RRGGen to Stage 4 output takes approximately six hours
1225 on a single GPU-equipped workstation. We should be precise about what
a reader can check without that run, because we audited it by execution
rather than assuming it. Re-running the released analysis scripts regenerates
roughly sixty stored result files bit-identically, including every statistical test
behind RQ1 and RQ2. What those scripts consume, however, is not itself
1230 regenerable from the release: the Stage-3 specification sets, the 100-cluster

sample, and the Stage-4 RAG sample ship as artifacts with no standalone producer script, since producing them requires the proprietary-API generation pass. The analysis layer is therefore reproducible from the middle outward, and the data-generation layer is reproducible only by re-running the pipeline against a paid API. One further artifact does not pin at all: the GitHub-mined human-written comparison set is re-queried live and returns a different issue mix on each run, so the `human_github` condition in Section 5.1 rests on a snapshot we can release but not regenerate. The release bundles:

- V1–V5 classifier checkpoints + cleanlab correction procedure;
- the 5,230-record verified anchor, which covers 5,229 distinct corpus rows and 490-review classification gold;
- both Stage-4 rating rounds, the discarded 400-row one and the reported 200-row re-run, together with the same-model generated replies and the rating-provenance check that distinguishes them;
- the count-controlled A1b output;
- the SPECCOV scorer plus the 60-spec, three-rater human faithfulness ratings and the validation analysis;
- the CMDP-RLHF testbed (KTO, DPO, constrained-proxy, and Lagrangian Constrained PPO trainers);
- the 100-cluster benchmark, with the 99-review LLM-assisted classification subsample ($\alpha = 0.285$) kept separate from the all-human agreement sets;
- a pinned-environment Docker image (single-command build on any CUDA 12.x host);
- a nineteen-segment verifier that recomputes the numerical claims in Section 5 from the saved data files, including the corrected Stage-4 re-run, the three-rater agreement statistics, the cross-family rubric, the SPECCOV validation, the binding-constraint run, and the generation-variance sweep. The discarded Stage-4 round retains its own segment, labelled as withdrawn. We audited the verifier itself rather than trusting it, and its coverage is narrower than we previously stated. Three segments recompute a value from the underlying data and assert it; the remainder load a stored summary and print it, which checks that the manuscript agrees with the released file but not that the file is right. Uncovered numbers include the Stage-3 template-fill table, the held-out κ of 0.616, the Qwen-judge purity figures, the Maalej cross-protocol row, the per-dimension cells of the cross-family rubric, and most of the secondary statistics in Sections 5.2 and 5.3. We list this rather than repeat the earlier and much smaller claim that

only two tables were uncovered. The verifier now exits non-zero on any mismatch, and every number in it that we could turn into an assertion we did; the rest a reader should check against the released files directly.

The pipeline is composable by construction: because each stage emits a typed intermediate, components (classifier, clusterer, taxonomy template, alignment trainer) can be substituted independently without breaking downstream contracts. Future researchers can adopt or replace any single component while keeping the rest of the IR-centered evaluation harness intact. Deterministic components (clustering, classifier training) use fixed seeds; the LLM-dependent stages (annotation, IR generation, and response generation) reproduce statistically rather than bit-identically. We measured how large that sampling variance is. Regenerating Stage 3 specs three times over the same 15 stratified clusters with the local Qwen2.5-3B generator at temperature 0.7, changing only the seed, gives strict template-fill 0.685, 0.700, and 0.634 (mean 0.673, sd 0.029, range 0.067) and SPEC COV 2.73, 2.67, and 3.00 (mean 2.80, sd 0.144, range 0.333), with no JSON parse failures in any run. Run-to-run noise on template-fill is therefore around three points on a 0–1 scale for this generator and prompt. That is the yardstick a reader should hold up against the differences we report. The gaps the paper rests on clear it comfortably: 0.96 versus 0.69 for taxonomy grounding, and 0.96 versus 0.48 across the model families. The withdrawn Stage-4 effect illustrates the opposite case, and is worth stating in these terms: at +0.04 it sits an order of magnitude below this noise floor, which is another way of saying there is nothing there to measure. Differences of a few hundredths between adjacent conditions, such as the 4.09 versus 3.96 at the top of the cross-family rubric, do not clear it and should not be read as a stable ordering on their own. Two limits remain: the headline Claude generations were not repeated, because that needs API budget rather than new code, and Stage 4 response generation was not repeated either. The released script runs both sweeps unchanged.

6.5. Future Work

Five extensions map directly to the limitations above; each starts from a resource released with this paper.

(0) Replace SPEC COV and widen the cross-family comparison.

The SPEC COV validation in Section 5.1 came back negative, so Stage 3 needs a faithfulness measure that actually works. The natural replacement is entailment-based: score each spec field against the source reviews with

1305 an NLI model [55, 56] or a question-generation-and-answering pipeline [57],
then validate that measure against the same 60-spec human ratings we release
here, which now serve as a ready-made benchmark. Second, the cross-family
rubric comparison (Table 8) rests on 14 matched bug-report clusters for the
four-way panel, because that is the extent of the overlap between the four
1310 generators’ spec sets, and it is scored by a judge from one of the four families.
Generating the missing spec types for the two Qwen models and re-scoring
with a judge outside all four families would remove both limits. Both are
runs rather than redesigns.

(1) **Ask the right raters, then scale.** The Stage 4 re-run carries three
1315 annotators on all 200 rows and returns a null with the raters disagreeing in
direction. Before growing the sample, the design should change: the reply
from a triage pipeline is consumed by a developer deciding what to do, not
by a general reader, so the next round should recruit raters with software-
engineering experience and no stake in the system. Scaling from 100 clusters
1320 to 200–300, with three annotators per IssueSpec and Krippendorff’s α at the
IssueSpec level, follows that rather than precedes it.

(2) **CMDP RLHF at a scale where the trade-off shows.** The
binding-constraint re-run in Section 5.3 establishes that the dual objective
engages; what it does not show is a trade-off, because a 5% base violation
1325 rate over 30 steps supplies too little signal. The next run wants a longer
horizon and a policy with a higher base violation rate, on an 8B-class model.
Original plan. Where the compliance constraint is expected to bind, activat-
ing the Lagrange multiplier and making the Pareto-dominance comparison
meaningful.

1330 (3) **Frontier-family replication.** GPT-4o and Gemini-1.5-Pro
on Stages 3–4. The current paper closes the Meta-family slice with
Llama-3.3-70B and the Alibaba slice with Qwen2.5-{3B, 1.5B}.

(4) **Cross-corpus transfer.** Stages 2–4 on additional review corpora;
V5 already transfers zero-shot to Maalej [1].

1335 (5) **Agentic Planner extension to Stage 4.** The Planner dispatches
a different sub-agent profile per issue type: defect-repair for `bug_report`
via SWE-Agent [11] or RepairAgent [12], scaffolding for `feature_request`,
profiler-driven optimization for `performance`, a Nielsen-heuristic checker for
`usability`, and a device-OS test-matrix executor for `compatibility`. A
1340 spec-level stub already runs the dispatch across all five issue types; end-to-
end patch evaluation on real repositories through HyperAgent [30], measured
by patch acceptance and test-pass rate, is the next step.

7. Conclusion

We proposed **IssueSpec**, a knowledge-graph-grounded typed intermediate representation that closes the five existing research gaps identified in Section 1: classification stops at the label, response generation skips the structured intermediate, agentic software engineering assumes the spec exists, knowledge graphs emit no typed schema, and RLHF has not addressed dev-rel compliance. We set out to show that the typed IR is the load-bearing component of spec-aware response generation, and we cannot. An isolation experiment that holds the generator fixed moves human-rated quality by +0.04 Likert points ($p = 0.38$, three raters, 100 paired reviews), and the raters disagree in direction. The +2.35 we previously reported came from a comparison whose two arms were written by different template composers. Nor does the negative half of our original hypothesis survive. We had reported that retrieval alone falls below a no-context baseline; against the arm that actually has no context it scores 2.24 against 2.31 at $p = 0.90$, and that comparison sits inside the discarded round in any case. Stage 4 yields one usable result, which is the null. A verified-anchor procedure makes the IR producible at corpus scale from noisy LLM labels, lifting Cohen’s κ from 0.163 to 0.592, and to 0.616 on the strictly held-out gold, with 88.66% classifier endorsement of corrections. Cross-family replication on four LLMs (Claude, Llama, Qwen-3B, Qwen-1.5B) preserves the rank order, ruling out single-model bias. Two findings travel beyond app reviews: an LLM labeller on software-engineering text can omit whole classes rather than merely err on them, and that a structured intermediate can be demonstrably better as an artifact while leaving the text generated from it indistinguishable, a distinction that pipeline evaluations of this shape should measure separately. We release the IssueSpec schema, the verified anchor, the SPECCOV scorer together with the human ratings that show it does not measure faithfulness, the count-controlled cluster-validation ablation, the structured-RAG architecture, the CMDP-RLHF trainers, and the benchmark dataset to support follow-on research.

GenAI Usage Disclosure

We disclose all uses of generative AI in this work in line with Elsevier policy on the use of generative AI in scientific writing. *Writing and code assistance*. An earlier version of this disclosure said that Claude assisted only

with grammatical refinement, caption polishing, and LaTeX formatting, and that all pipeline code, experimental scripts, and analysis were written by the human authors. That understated the use, and the repository we release contradicts it: 38 of the 47 commits carry a **Co-Authored-By: Claude** trailer, including commits that add pipeline stages, the constrained-PPO trainer, the analysis scripts, and the correction passes described in Section 5.2. The accurate statement is that Anthropic’s Claude was used throughout implementation and analysis as a coding assistant, under author direction and review, and also for manuscript editing. Every design decision, every experimental choice, and every claim in this paper is the authors’ own, and the numbers were produced by running the released code on the released data. We prefer to correct this here rather than leave a disclosure that the artifact refutes. Every structural, framing, experimental-design, and claim decision was made by the authors, and every reported number was produced by author-written code run under human supervision and verified against saved data files. *Models inside the system.* Several generative models are themselves pipeline components and are reported as such in the body: Claude (Opus 4.7) is the headline Stage 3 generator (Table 11); Llama-3.3-70B-Instruct (via Groq) and Qwen2.5-{3B, 1.5B}-Instruct are cross-family Stage 3 comparators (Table 9); Qwen2.5-3B-Instruct is the LLM-as-judge for the five-dimension rubric and the Y/P/N cluster-purity audit, both calibrated against the lead author (Tables 10, 15). The V2 corpus labels in Section 5.1 are LLM-generated and explicitly treated as noisy, which is precisely what motivates the verified-anchor procedure of Stage 1. *Human verification.* No reported numerical result, expert annotation, or ground-truth label was automatically generated by AI without human verification. The 5,230-review verified anchor, the 490-review classification gold, the 99-review three-rater purity ratings, and the 20 lead-author IssueSpec references were all produced and reviewed by human annotators (Section 4).

CRediT authorship contribution statement

Fabiha Jalal: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. **Sadia Tasnim Dhruba:** Investigation, Data curation, Validation, Writing – review and editing. **Tazrian Zaman Tushti:** Investigation, Data curation, Validation, Writing – review and editing. **Ajwad Abrar:** Methodology, Supervision, Writing – review

and editing. **Md Kamrul Hasan:** Supervision, Resources, Writing – review
1415 and editing. **Hasan Mahmud:** Supervision, Resources, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work
1420 reported in this paper.

Data availability

All code, configuration files, and analysis scripts are available in the project repository. A verification bundle that reproduces every numerical claim in this paper is archived on Zenodo (DOI 10.5281/zenodo.21982774, v2.0), and the Stage-1 V5 classifier is released on Hugging Face. That bundle
1425 supersedes an earlier release (DOI 10.5281/zenodo.20320410) which predates the Stage-4 correction described in Section 5.2; both versions remain resolvable, and the earlier one should not be used. The underlying review corpus derives from the publicly released RRGGen dataset of Gao et al.; derived artifacts are redistributed subject to the terms of that original release.
1430

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Appendix A. Implementation Settings

Section 3 states what each stage does and why. This appendix collects the exact settings, so the method section carries the logic and the numbers stay checkable. Stage 1 fine-tunes a RoBERTa multi-label head; confident-learning correction against the verified anchor is gated by anchor confidence ≥ 0.70 and LLM probability ≤ 0.20 , with a 3×3 sensitivity grid over $(0.65, 0.70, 0.75) \times (0.15, 0.20, 0.25)$ reported in Section 5.1. The `compatibility` stratum is seeded with 200 template-generated synthetic reviews (device, OS, and screen templates) plus 100 keyword-mined RRGGen reviews, all human-checked. Stage 2 embeds with Sentence-BERT, reduces with UMAP, and clusters with HDBSCAN, aspect-first, producing 605 sub-clusters of mean size 15.9 against 194 flat clusters of mean size 471.0. The KG-hierarchical side runs over an 8,404-review sample rather than the full corpus; the flat baseline runs over 91,368 clustered reviews drawn from the full corpus, so the two sides describe different review sets (Section 5). PageRank runs on the bipartite review-aspect graph. Stage 3 routes to one of five templates and re-prompts once with dimension-level feedback on any rubric failure. Stage 4 retrieves over a 15,100-document ChromaDB index built from five sources, with the IssueSpec injected per query rather than pre-indexed. Stage 5 trains five policies on distilGPT2 (82M parameters) from 400 SFT

samples, 296 KTO pairs, and 100 DPO pairs; the Lagrangian loop runs 30 steps at batch size 4 with KL coefficient 0.05, and the compliance threshold sweep covers $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$. SPEC_{COV} counts substantive-token overlap with the source cluster, tokens of length ≥ 5 after stop-word filtering, with quoted-string and coverage bonuses.

Appendix B. Annotator Guidelines

The lead author (PhD in software engineering with prior mobile-app review-mining experience) served as primary annotator across all five annotation tasks. The 490-review classification gold was additionally annotated by two further independent human raters on the identical item set, aligned by review id; all three first completed a shared 20-review calibration round, disagreements were adjudicated before the main task opened, and the released gold label is the majority vote over the three raters. All human annotation in this work is carried out by those same three people. For Stage 4, all three raters scored all 200 rows of the reported re-run independently and blind to condition; the discarded 400-row round had the lead author as primary rater with one additional rater on a 30-pair subset, and is described here only because the paper reports its withdrawal. Separately, two Qwen2.5-3B-Instruct prompts (concise role-based and chain-of-thought) served as the second and third raters in the 99-review cluster-purity Krippendorff α calibration, which is an LLM-assisted panel and not one of the human rounds. For the 5,230-record verified anchor, which covers 5,229 distinct corpus rows, annotators classified each review into one of seven Maalej-Nabil classes (**bug_report**, **feature_request**, **performance**, **usability**, **compatibility**, **praise**, **other**) with multi-label assignment permitted when a review carried multiple concerns; reviews below the “fairly confident” threshold were skipped and excluded from the anchor. For the 490-review classification gold standard, a stratified sample of 70 reviews per class was drawn from the working corpus, and the lead author re-labeled each review independently and blinded to the V2 LLM prediction; the 490 gold labels serve as the held-out evaluation set for Cohen’s κ progression (Table 12) and per-class F1 (Table 13). For the 5-dimension IssueSpec rubric (all 100 specs; an earlier stratified $n = 28$ subsample, 6 per issue type plus 2 extras), each spec was scored on a 1–5 Likert scale across five dimensions: *completeness* (are all required template fields filled?), *accuracy* (do field values correctly reflect the source cluster?), *actionability* (would a developer

know what to do?), *specificity* (does the spec name concrete entities rather than generic terms?), and *clarity* (is the spec easy to read and unambiguous?), with each dimension scored independently. For the reported Stage 4 evaluation (100 reviews $\times$ 2 conditions, the same model with and without the IssueSpec, 200 rows), all three raters rated *quality* (1–5), *specificity* (1–5), and *helpful* (Y/N) while blinded to the generation method; the superseded 400-row round used four template-composed conditions and is retained in the release for comparison. In both rounds condition labels were replaced with random A/B/C/D codes and unsealed only at aggregation, with the assignment persisted as spreadsheets with JSON master keys for audit. A second human rater independently scored a 30-pair (`no_spec`, `full`) subset for inter-rater calibration. For the Y/P/N cluster-purity rubric (initial 50-cluster audit; 99-cluster three-rater subsample), annotators read 5 representative reviews per cluster and assigned *Y* (all 5 share the same sub-theme within the cluster’s aspect), *P* (3 or 4 of 5 share the sub-theme), or *N* (fewer than 3 share); weighted purity is $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$, with *Y* contributing 1.0, *P* contributing 0.5, and *N* contributing 0. All comparative evaluations were blinded throughout the rating window and unsealed only at aggregation; reviews or specs below the annotator confidence threshold were marked as “skip” and excluded from downstream analysis.

Table 6: Annotator-vs-LLM and inter-annotator deviation. The $\approx 25\%$ deviation is measured on the human-verified anchor; the κ -progression reaches moderate agreement [51]; the V5 endorsement row is a consistency check, not independent validation, since V5 trained on those corrections; LLM-judge calibration Δ bounds judge bias.

Annotator pair	sample	agreement / deviation
Lead-author vs V2 LLM	5,230	75% / 25% mislabeled
Lead vs V2 / cleanlab / V5	489 gold	κ : 0.163 $\rightarrow$ 0.333 $\rightarrow$ 0.592
3 human raters, Y/N verdict	490 gold	Fleiss $\kappa =$ 0.968 ; 479/490 unanimous on the verdict, 476/489 on the label
3 human raters, 7-class label	489 gold	Fleiss $\kappa =$ 0.979 ; $\alpha = 0.979$
V2 LLM vs V5; cleanlab vs V5	215,583	71.96% / 86.77% agree
cleanlab vs V5 (consistency, not independent)	40,291	88.66% support
V5 vs gold, strictly held out	307	$\kappa =$ 0.616
3 human raters (Stage 4 quality)	400 rated rows	weighted κ 0.970–0.986; within-1 100%
3 human raters (Stage 4 helpful)	400 rated rows	κ 0.678–0.836; $\alpha = 0.781$
3-rater <i>LLM-assisted</i> classification check (lead + 2 Qwen prompts)	99 re-views	Krippendorff $\alpha = 0.285$
Lead vs Qwen-judge (purity, rubric)	50 / 28	$\Delta = -0.035 / +0.23$

Table 7: What the experiments establish, in one line each. Negative findings are marked, and they are findings rather than caveats.

	finding	evidence
F1	Type routing, not the language model, is what produces tracker-ready fields: removing it collapses strict fill from 0.96 to 0.69 and the per-type structural metrics to zero.	§5.1, A3
F2	<i>Corrected.</i> We reported that LLM annotation noise concentrates on minority classes. Per stratum it does the opposite: 1,302 of 1,307 disagreements are in the largest class. What holds is that the labeller omits two classes entirely, which is the failure a verified anchor repairs.	§5.1
F3	<i>Negative, and it overturns our own earlier claim.</i> With the generator held fixed, adding the typed IR changes response quality by +0.04 ($p = 0.38$); the +2.35 we previously reported was an artifact of comparing two different text composers.	§5.2
F4	Generator capability shows up mostly as whether fields get filled, not as the quality of the text once filled: template-fill spans 0.96–0.48 across four models while rubric quality spans 4.09–3.77.	§5.1
F5	<i>Negative.</i> The knowledge graph loses to count-matched flat clustering on all three intrinsic metrics; its defensible role is centrality-based prioritization.	§5.4, A1b
F6	<i>Negative.</i> Extractive coverage does not measure faithfulness: SPEC _{COV} shows no rank correlation with human judgement and orders generators backwards.	§5.1
F7	<i>Negative, then partial.</i> A CMDP constraint cannot be tested on a policy incapable of violating it; with one that can, the constraint binds and λ grows, but no trade-off is demonstrated.	§5.3
F8	Run-to-run generation noise is about 0.03 on template-fill, which is the yardstick separating the differences we claim from the ones we do not.	§6.4

Table 8: Cross-family five-dimension rubric, same Qwen2.5-3B judge as Table 10, on matched clusters. Panel A is the four-way comparison on clusters covered by every generator (bug reports only). Panel B widens the Claude-versus-Llama comparison to all five issue types. The Panel-A ordering matches the template-fill ordering in Table 9, but none of its six pairwise differences is significant at $n = 14$ (Wilcoxon p from 0.18 to 0.75), so read it as descriptive. Panel B, with twice the sample and all five issue types, does separate: +0.38 for Claude over Llama, Wilcoxon $p = 0.015$.

generator	n	mean	compl.	accur.	action.	spec.	clarity
<i>Panel A, four-way, 14 matched bug-report clusters</i>							
Claude (Anthropic)	14	4.09	4.00	4.50	3.71	4.36	3.86
Llama-3.3-70B (Meta)	14	3.96	3.86	4.50	4.07	3.86	3.50
Qwen2.5-3B (Alibaba)	14	3.83	3.07	4.50	4.00	3.86	3.71
Qwen2.5-1.5B (Alibaba)	13	3.77	3.00	4.15	4.46	3.62	3.62
<i>Panel B, Claude vs Llama, 28 clusters, all five issue types</i>							
Claude (Anthropic)	28	4.01	3.86	4.54	3.75	4.29	3.64
Llama-3.3-70B (Meta)	28	3.64	3.39	4.46	3.14	3.82	3.36

Table 9: Multi-LLM evidence summary across three model families (Anthropic, Meta, Alibaba). Cross-family rank preservation across Stage 3 and Stage 4 is the structural-bias control; LLM-as-judge calibration Δ vs the lead author bounds judge bias.

Use	Model (family)	Result
Stage 3 generate	Claude (Anthropic) / Llama-3.3-70B (Meta) / Qwen-3B (Alibaba) / Qwen-1.5B (Alibaba)	0.96 / 0.80 / 0.74 / 0.48
Stage 4 generate	Claude / Qwen-3B	0.61 / 0.54
5-dim rubric judge	Qwen-3B, all 100 specs	3.94/5 (28-spec subsample: 3.89)
Cross-family rubric	Claude / Llama / Qwen-3B / Qwen-1.5B	4.09 / 3.96 / 3.83 / 3.77 (Table 8)
Y/P/N judge	Qwen-3B vs lead-author	0.625 vs 0.66; $\Delta = -0.035$

Table 10: Experiment 1 five-dimension rubric (Qwen-judge, 1–5 scale) on the *complete* spec set: all 100 taxonomy specs and all 20 lead-author reference specs, scored in one run so the two columns are directly comparable. The middle column repeats the earlier stratified 28-spec subsample under the same judge and rubric; the gap between it and the full set is +0.05, which is the subsample error. The LLM-as-judge protocol follows the calibration-against-reference convention of [44, 45, 46].

dimension	taxonomy, $n = 100$	taxonomy, $n = 28$	lead-author ref, $n = 20$
completeness	3.72	3.79	3.20
accuracy	4.43	4.39	4.10
actionability	3.86	3.71	2.85
specificity	4.15	4.07	4.05
clarity	3.52	3.50	3.05
overall mean	3.94	3.89	3.45
sd across specs	0.59	—	0.41

Table 11: Experiment 1 substantive template-fill (strict criteria). The 0.96 vs 0.53 LLM-vs-GitHub gap is coverage, not content quality; the strict-vs-very-strict sweep preserves ordering. The exact-zero cells in **bugs:repro** and **features:story** columns for (b), (c), (d), and (e) are structural: free-form LLM, raw concatenation, lead-author, and human-GitHub specs do not emit the template-specific reproduction-step list or As-a/I-want/So-that triple at all when not prompted for them, so the strict-fill rate is zero by construction.

condition	strict	bugs:repro	features:story
(a) LLM + taxonomy	0.96	0.73	0.97
(b) LLM free-form	0.69	0.00	0.00
(c) raw concat	0.34	0.00	0.00
(d) lead-author ref ($n = 20$)	0.69	0.00	0.00
(e) human GitHub (3 repos)	0.53	0.13	0.00

Table 12: Cohen’s κ progression vs the 490-review expert gold. Three Landis-Koch [51] bands crossed; cleanlab correction follows the confident-learning procedure of [23]. The final column repeats the measurement against the majority-vote gold of the three-rater human panel, showing the trajectory is not an artifact of a single annotator.

classifier	n	Cohen’s κ	Landis-Koch band	κ vs 3-rater gold
V2 LLM original	489	0.163	slight	0.165
cleanlab-corrected V2	489	0.333	fair	0.334
V5 classifier	489	0.592	moderate	0.590

Table 13: Per-class F1 against the 490-review expert gold. V2 LLM is blind to the two minority classes (compatibility, performance). The verified-anchor procedure with targeted augmentation recovers them.

class	V2 LLM	cleanlab-V2	V5
bug_report	0.42	0.45	0.58
feature_request	0.32	0.37	0.57
performance	0.00	0.47	0.77
usability	0.11	0.27	0.55
compatibility	0.00	0.00	0.83
praise	0.38	0.62	0.68
other	0.30	0.47	0.60
macro F1	0.22	0.38	0.65

Table 14: Stage-4 re-run, same model with and without the IssueSpec, 100 paired reviews, three raters, all 200 rows scored by each. All figures are unadjusted, so the gain column equals the difference of the two condition columns; slot-stratified values are in the released JSON and differ by at most 0.03. The pooled effect is null, and the disagreement between the lead author and the two independent raters is larger than the effect being measured.

rater	no_spec	full	gain	95% CI	Wilcoxon p
lead author	4.04	4.60	+0.56	[+0.44, +0.68]	2×10^{-11}
independent rater 1	3.87	3.58	-0.29	[-0.56, -0.01]	0.06
independent rater 2	3.57	3.42	-0.15	[-0.33, +0.03]	0.09
pooled	3.83	3.87	+0.04	[-0.10, +0.18]	0.38

Table 15: Stage 2 designs head-to-head, comparing Pure HDBSCAN (without UMAP), original flat UMAP+HDBSCAN, and KG-hierarchical. Intrinsic metrics (Davies-Bouldin [58], Calinski-Harabasz [59], silhouette [60]) follow standard IR/clustering convention [61, 62, 63]. The Y/P/N weighted-purity rubric is a discrete-coding instantiation of the extrinsic cluster-purity metric [64, 61]. The count-controlled flat baseline (agglomerative-605) is reported separately in the A1b ablation (§5.5).

metric	Pure HDB	Flat U+H	KG hier.
n_clusters	3 (deg.)	194	605
mean cluster size	~16,933	471.0	15.9
Davies-Bouldin ($\downarrow$)	n/a	<i>withdrawn</i>	2.24
Calinski-Harabasz ($\uparrow$)	n/a	<i>withdrawn</i>	1.85
Y/P/N purity (lead-author)	n/a	0.66	queued
Y/P/N purity (Qwen-judge)	n/a	0.625	0.390

Table 16: Ablation summary. “Removing this hurts” is the only outcome that justifies a component. Three of the seven do not clear that bar, and we report them as findings.

id	what is removed	effect of removing it	verdict
A1	the KG, feeding flat clusters to Stage 3	Stage 3 quality unchanged	no effect
A1b	the KG, against flat clustering matched to the same 605 clusters	flat wins on all three intrinsic metrics	against the KG
A2	hierarchical sub-clustering	purity 0.625 flat vs 0.390 KG	against the KG
A3	taxonomy grounding, so the LLM writes free-form	strict fill 0.96 $\rightarrow$ 0.69; per-type structure to zero	justifies routing
A4	HITL-2 review of specs	no change to benchmark scores	no effect at benchmark time
A5	retrieval, keeping the IssueSpec	all automatic metrics move $< 1\%$	against RAG
A6	the IssueSpec, keeping retrieval	quality 3.87 $\rightarrow$ 3.83 ($-0.04, p = 0.38$)	no measurable effect (§5.2)
A7	single-stream feedback	RLHF not run	deferred